

kohlhase-omdoc: formula evidence

The page, the confidence and the picture of an inline formula are its HOST LINE’s — a formula has none of its own. A line’s confidence is not a formula’s.

Inline formulas (first occurrence)

| Identifier            | Page | Conf.   | LaTeX source                                                                                    | Rendered                                                                                     | Image                                                                                                                                                                                           |
|-----------------------|------|---------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| kohlhase-omdoc_F00001 | 10   | ■ 1.000 | \Omega                                                                                          | $\Omega$                                                                                     | tably the $\Omega$ MEGA group at Saarland University, the MAYA and ACTIVEMATH                                                                                                                   |
| kohlhase-omdoc_F00002 | 19   | ■ 0.928 | \mathrm{T}_{\mathrm{E}} \mathrm{X} / \mathrm{LA}_{\mathrm{E}} \mathrm{X}                        | $\mathrm{T}_{\mathrm{E}}\mathrm{X}/\mathrm{LA}_{\mathrm{E}}\mathrm{X}$                       | distinguish them from presentation markup schemes, and discuss $\mathrm{T}_{\mathrm{E}}\mathrm{X}/\mathrm{L}^{\mathrm{A}}\mathrm{T}_{\mathrm{E}}\mathrm{X}$                                     |
| kohlhase-omdoc_F00003 | 19   | ■ 1.000 | \mathrm{T}_{\mathrm{E}} \mathrm{X}                                                              | $\mathrm{T}_{\mathrm{E}}\mathrm{X}$                                                          | $\mathrm{T}_{\mathrm{E}}\mathrm{X}$ is a typesetting markup language that uses explicit markup codes                                                                                            |
| kohlhase-omdoc_F00004 | 19   | —       | \sqrt{\sin x}\backslash                                                                         | (not rendered)                                                                               | —                                                                                                                                                                                               |
| kohlhase-omdoc_F00005 | 19   | ■ 0.878 | \sqrt{\sin x}                                                                                   | $\sqrt{\sin x}$                                                                              | $\sqrt{\sin x}$ stands for the mathematical expression $\sqrt{\sin x}$ in $\mathrm{T}_{\mathrm{E}}\mathrm{X}$ . To                                                                              |
| kohlhase-omdoc_F00006 | 21   | ■ 0.696 | \mathrm{T}_{\mathrm{E}} \mathrm{X} / \mathrm{La}_{\mathrm{T}} \mathrm{E} \mathrm{X}             | $\mathrm{T}_{\mathrm{E}}\mathrm{X}/\mathrm{La}_{\mathrm{T}}\mathrm{E}\mathrm{X}$             | to $\mathrm{T}_{\mathrm{E}}\mathrm{X}/\mathrm{La}_{\mathrm{T}}\mathrm{E}\mathrm{X}$ in spirit, but tries to give the markup scheme a more declar-                                               |
| kohlhase-omdoc_F00007 | 23   | ■ 0.940 | \langle\!\!\langle nsURI \rangle\!\!\rangle                                                     | $\langle\!\!\langle nsURI \rangle\!\!\rangle$                                                | descendants are in the namespace $\langle\!\!\langle nsURI \rangle\!\!\rangle$ , unless they have a namespace                                                                                   |
| kohlhase-omdoc_F00008 | 23   | ■ 0.198 | \langle                                                                                         | $\langle$                                                                                    | by a pseudo-attribute of the form $\mathrm{xmlns}:\langle\!\!\langle ns \rangle\!\!\rangle=\langle\!\!\langle nsUR \rangle\!\!\rangle$ , where $\langle\!\!\langle ns \rangle\!\!\rangle$ is    |
| kohlhase-omdoc_F00009 | 23   | ■ 0.198 | \rangle=                                                                                        | $\rangle=$                                                                                   | by a pseudo-attribute of the form $\mathrm{xmlns}:\langle\!\!\langle ns \rangle\!\!\rangle=\langle\!\!\langle nsUR \rangle\!\!\rangle$ , where $\langle\!\!\langle ns \rangle\!\!\rangle$ is    |
| kohlhase-omdoc_F00010 | 23   | ■ 0.198 | \rangle                                                                                         | $\rangle$                                                                                    | by a pseudo-attribute of the form $\mathrm{xmlns}:\langle\!\!\langle ns \rangle\!\!\rangle=\langle\!\!\langle nsUR \rangle\!\!\rangle$ , where $\langle\!\!\langle ns \rangle\!\!\rangle$ is    |
| kohlhase-omdoc_F00011 | 23   | ■ 0.198 | \langle\!\!\langle                                                                              | $\langle\!\!\langle$                                                                         | by a pseudo-attribute of the form $\mathrm{xmlns}:\langle\!\!\langle ns \rangle\!\!\rangle=\langle\!\!\langle nsUR \rangle\!\!\rangle$ , where $\langle\!\!\langle ns \rangle\!\!\rangle$ is    |
| kohlhase-omdoc_F00012 | 23   | ■ 0.198 | \rangle\!\!\rangle                                                                              | $\rangle\!\!\rangle$                                                                         | by a pseudo-attribute of the form $\mathrm{xmlns}:\langle\!\!\langle ns \rangle\!\!\rangle=\langle\!\!\langle nsUR \rangle\!\!\rangle$ , where $\langle\!\!\langle ns \rangle\!\!\rangle$ is    |
| kohlhase-omdoc_F00013 | 23   | ■ 0.190 | nsa)\rangle                                                                                     | $nsa)\rangle$                                                                                | a <b>qualified name</b> : a pair $\langle\!\!\langle ns \rangle\!\!\rangle:\langle\!\!\langle el \rangle\!\!\rangle$ , where $\langle\!\!\langle ns \rangle\!\!\rangle$ is a namespace abbrevi- |
| kohlhase-omdoc_F00014 | 23   | ■ 0.190 | nsa\rangle                                                                                      | $nsa\rangle$                                                                                 | a <b>qualified name</b> : a pair $\langle\!\!\langle ns \rangle\!\!\rangle:\langle\!\!\langle el \rangle\!\!\rangle$ , where $\langle\!\!\langle ns \rangle\!\!\rangle$ is a namespace abbrevi- |
| kohlhase-omdoc_F00015 | 23   | ■ 0.917 | \hat{=}                                                                                         | $\hat{=}$                                                                                    | $\langle\!\!\langle CDATA[a<b<\sup>3</sup>]>\rangle\!\!\rangle \hat{=} a\&lt;b\&lt;\sup\&gt;3\&lt;/\sup\&gt;$                                                                                   |
| kohlhase-omdoc_F00016 | 25   | ■ 0.832 | 7^{\text{th}}                                                                                   | $7^{\mathrm{th}}$                                                                            | the $7^{\mathrm{th}}$ child of the $3^{\mathrm{rd}}$ child of the (unique) element that has ID-type attribute                                                                                   |
| kohlhase-omdoc_F00017 | 25   | ■ 0.832 | 3^{\text{rd}}                                                                                   | $3^{\mathrm{rd}}$                                                                            | the $7^{\mathrm{th}}$ child of the $3^{\mathrm{rd}}$ child of the (unique) element that has ID-type attribute                                                                                   |
| kohlhase-omdoc_F00018 | 25   | ■ 0.984 | D                                                                                               | $D$                                                                                          | (the shorthand <b>xpointer</b> ): If $\langle\!\!\langle uri \rangle\!\!\rangle$ is a URI of an XML document $D$ then                                                                           |
| kohlhase-omdoc_F00019 | 25   | ■ 0.454 | \langle\!\!\langle id \rangle\!\!\rangle                                                        | $\langle\!\!\langle id \rangle\!\!\rangle$                                                   | ID with value $\langle\!\!\langle id \rangle\!\!\rangle$ .                                                                                                                                      |
| kohlhase-omdoc_F00020 | 27   | ■ 0.990 | \mathrm{T}_{\mathrm{E}} \mathrm{X} / \mathrm{L}^{\mathrm{A}} \mathrm{T}_{\mathrm{E}} \mathrm{X} | $\mathrm{T}_{\mathrm{E}}\mathrm{X}/\mathrm{L}^{\mathrm{A}}\mathrm{T}_{\mathrm{E}}\mathrm{X}$ | systems like $\mathrm{T}_{\mathrm{E}}\mathrm{X}/\mathrm{L}^{\mathrm{A}}\mathrm{T}_{\mathrm{E}}\mathrm{X}$ .                                                                                     |
| kohlhase-omdoc_F00021 | 28   | ■ 0.348 | { }^{+}                                                                                         | $+$                                                                                          | formulae for the Web are MATHML [ABC <sup>+</sup> 03] and OPENMATH [BCC <sup>+</sup> 04].                                                                                                       |
| kohlhase-omdoc_F00022 | 28   | ■ 1.000 | { }^{\text{®}}                                                                                  | $\text{®}$                                                                                   | mathematical software packages like Wolfram Research’s MATHEMATICA <sup>®</sup>                                                                                                                 |
| kohlhase-omdoc_F00023 | 29   | ■ 1.000 | \frac{3}{x+2}                                                                                   | $\frac{3}{x+2}$                                                                              | the expression $\frac{3}{x+2}$ would be represented as the layout tree in Figure 2.2.                                                                                                           |

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| kohlhase-omdoc_<br>F00024 | 29   | ■ 0.991 | \Sigma                                      | $\Sigma$                                                  | DTD, e.g. <code>&amp;sum</code> ; for $\Sigma$ .                                                                                                                 |
| kohlhase-omdoc_<br>F00025 | 29   | ■ 0.840 | 14{ }^{\text{th}}                           | 14 <sup>th</sup>                                          | fixed set of mathematical concepts – the K-14 fragment (Kindergarten to 14 <sup>th</sup>                                                                         |
| kohlhase-omdoc_<br>F00026 | 30   | ■ 0.998 | \mathrm{T}_{\mathrm{E}} \mathrm{X}^3        | $\mathrm{T}_\mathrm{E}X^3$                                | of $\mathrm{T}_\mathrm{E}X$ . However, it does not offer the programming language features of $\mathrm{T}_\mathrm{E}X^3$                                         |
| kohlhase-omdoc_<br>F00027 | 30   | ■ 1.000 | { }^3 \mathrm{~T}_{\mathrm{E}} \mathrm{X}   | ${}^3 \mathrm{T}_\mathrm{E}X$                             | ${}^3 \mathrm{T}_\mathrm{E}X$ contains a full, Turing-complete – if somewhat awkward – programming lan-                                                          |
| kohlhase-omdoc_<br>F00028 | 33   | ■ 1.000 | \forall a, b . a+b=b+a                      | $\forall a, b. a + b = b + a$                             | (the formula $\forall a, b. a + b = b + a$ ) in OPENMATH. Note that as we have dis-                                                                              |
| kohlhase-omdoc_<br>F00029 | 35   | ■ 1.000 | S=(G, \circ)                                | $S = (G, \circ)$                                          | A monoid is a semigroup $S = (G, \circ)$ with an element $e \in G$ , such that                                                                                   |
| kohlhase-omdoc_<br>F00030 | 35   | ■ 1.000 | e \in G                                     | $e \in G$                                                 | A monoid is a semigroup $S = (G, \circ)$ with an element $e \in G$ , such that                                                                                   |
| kohlhase-omdoc_<br>F00031 | 35   | ■ 0.988 | e \circ x=x                                 | $e \circ x = x$                                           | $e \circ x = x$ for all $x \in G$ . $e$ is called a left unit of $S$ .                                                                                           |
| kohlhase-omdoc_<br>F00032 | 35   | ■ 0.988 | x \in G                                     | $x \in G$                                                 | $e \circ x = x$ for all $x \in G$ . $e$ is called a left unit of $S$ .                                                                                           |
| kohlhase-omdoc_<br>F00033 | 35   | ■ 0.988 | e                                           | $e$                                                       | $e \circ x = x$ for all $x \in G$ . $e$ is called a left unit of $S$ .                                                                                           |
| kohlhase-omdoc_<br>F00034 | 35   | ■ 0.988 | S                                           | $S$                                                       | $e \circ x = x$ for all $x \in G$ . $e$ is called a left unit of $S$ .                                                                                           |
| kohlhase-omdoc_<br>F00035 | 35   | ■ 0.915 | f                                           | $f$                                                       | <b>Proof:</b> We assume that there is another left unit $f \dots$                                                                                                |
| kohlhase-omdoc_<br>F00036 | 35   | —       | \square                                     | $\square$                                                 | $\square$                                                                                                                                                        |
| kohlhase-omdoc_<br>F00037 | 37   | ■ 0.491 | \mathrm{BCF}^+                              | $\mathrm{BCF}^+$                                          | MIZAR [Rud92] and $\Omega$ MEGA [BCF <sup>+</sup> 97] which utilize strong logical systems                                                                       |
| kohlhase-omdoc_<br>F00038 | 38   | ■ 0.995 | \left[\mathrm{SBB}^+{}_{02}\right]          | [SBB <sup>+</sup> 02]                                     | on the other. $\Omega$ MEGA [SBB <sup>+</sup> 02] is a large-scale proof development environ-                                                                    |
| kohlhase-omdoc_<br>F00039 | 40   | ■ 0.772 | 1{ }^{\text{st}} 2000                       | 1 <sup>st</sup> 2000                                      | Version 1.0 of the OMDoc format was released on November 1 <sup>st</sup> 2000 to                                                                                 |
| kohlhase-omdoc_<br>F00040 | 40   | ■ 0.981 | 29{ }^{\text{th}} 2001                      | 29 <sup>th</sup> 2001                                     | OMDoc 1.1 was released on December 29 <sup>th</sup> 2001 as an attempt to roll                                                                                   |
| kohlhase-omdoc_<br>F00041 | 41   | ■ 0.996 | X+0=0</ \mathrm{FMP}>                       | $X + 0 = 0 < / \mathrm{FMP} >$                            | <FMP> $X + 0 = 0$ </FMP>                                                                                                                                         |
| kohlhase-omdoc_<br>F00042 | 41   | ■ 0.999 | <\mathrm{FMP}>X+s(Y)=s(X+Y)</ \mathrm{FMP}> | $< \mathrm{FMP} > X + s(Y) = s(X + Y) < / \mathrm{FMP} >$ | <FMP> $X + s(Y) = s(X + Y)$ </FMP>                                                                                                                               |
| kohlhase-omdoc_<br>F00043 | 42   | ■ 0.712 | { }^+{}_{04}                                | <sup>+</sup> 04                                           | mulae, OMDoc uses the established standards OPENMATH [BCC <sup>+</sup> 04]                                                                                       |
| kohlhase-omdoc_<br>F00044 | 43   | ■ 0.501 | { }^{\text{TM}}                             | <sup>TM</sup>                                             | On the left hand side we see CAS like MATHEMATICA <sup>®</sup> [Wol02] or MAPLE <sup>™</sup>                                                                     |
| kohlhase-omdoc_<br>F00045 | 43   | ■ 0.820 | \left[\mathrm{CGG}^+{}_{92}\right]          | [CGG <sup>+</sup> 92]                                     | [CGG <sup>+</sup> 92] that are relatively restricted in the mathematical objects — they                                                                          |
| kohlhase-omdoc_<br>F00046 | 48   | ■ 0.988 | \mathrm{LAT}_{\mathrm{E}} \mathrm{X}        | $\mathrm{LAT}_\mathrm{E}X$                                | the $\mathrm{T}_\mathrm{E}X/\mathrm{L}\mathrm{T}_\mathrm{E}X$ system, we will take a $\mathrm{L}\mathrm{A}\mathrm{T}_\mathrm{E}X$ representation as the starting |
| kohlhase-omdoc_<br>F00047 | 49   | ■ 0.851 | E                                           | $E$                                                       | DEFINITION 1. Let $E$ be a set. A mapping of $E \times E$ is called a law of composition                                                                         |
| kohlhase-omdoc_<br>F00048 | 49   | ■ 0.851 | E \times E                                  | $E \times E$                                              | DEFINITION 1. Let $E$ be a set. A mapping of $E \times E$ is called a law of composition                                                                         |
| kohlhase-omdoc_<br>F00049 | 49   | ■ 0.988 | f(x, y)                                     | $f(x, y)$                                                 | on $E$ . The value $f(x, y)$ of $f$ for an ordered pair $(x, y) \in E \times E$ is called the                                                                    |
| kohlhase-omdoc_<br>F00050 | 49   | ■ 0.988 | (x, y) \in E \times E                       | $(x, y) \in E \times E$                                   | on $E$ . The value $f(x, y)$ of $f$ for an ordered pair $(x, y) \in E \times E$ is called the                                                                    |
| kohlhase-omdoc_<br>F00051 | 49   | ■ 0.965 | x                                           | $x$                                                       | composition of $x$ and $y$ under this law. A set with a law of composition is called                                                                             |

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| kohlhase-omdoc_F00052 | 49   | ■ 0.965 | $y$                                                                                       | $y$                                                                         | composition of $x$ and $y$ under this law. A set with a law of composition is called                                                                       |
| kohlhase-omdoc_F00053 | 49   | ■ 0.999 | $x+y, x \cdot y$                                                                          | $x + y, x.y$                                                                | symbols the composition of $x$ and $y$ is written respectively as $x + y, x.y$ or $xy$ .                                                                   |
| kohlhase-omdoc_F00054 | 49   | ■ 0.999 | $x \ y$                                                                                   | $xy$                                                                        | symbols the composition of $x$ and $y$ is written respectively as $x + y, x.y$ or $xy$ .                                                                   |
| kohlhase-omdoc_F00055 | 49   | ■ 1.000 | $x+y$                                                                                     | $x + y$                                                                     | $x + y$ being called the <i>sum</i> of $x$ and $y$ ) and we say that it is <i>written additively</i> ;                                                     |
| kohlhase-omdoc_F00056 | 49   | ■ 0.999 | $x \cdot y = x \ y$                                                                       | $x.y = xy$                                                                  | <u><math>x.y = xy</math></u> being called the <i>product</i> for $x$ and $y$ ) and we say that it is <i>written</i>                                        |
| kohlhase-omdoc_F00057 | 49   | ■ 0.964 | $\top$                                                                                    | $\top$                                                                      | use the symbols $\top$ and $\perp$ to denote arbitrary laws of composition.                                                                                |
| kohlhase-omdoc_F00058 | 49   | ■ 1.000 | $(X, Y) \mapsto X \cup Y$                                                                 | $(X, Y) \mapsto X \cup Y$                                                   | <i>Examples.</i> (1) The mappings $(X, Y) \mapsto X \cup Y$ and $(X, Y) \mapsto X \cap Y$ are                                                              |
| kohlhase-omdoc_F00059 | 49   | ■ 1.000 | $(X, Y) \mapsto X \cap Y$                                                                 | $(X, Y) \mapsto X \cap Y$                                                   | <i>Examples.</i> (1) The mappings $(X, Y) \mapsto X \cup Y$ and $(X, Y) \mapsto X \cap Y$ are                                                              |
| kohlhase-omdoc_F00060 | 49   | ■ 1.000 | $x \in \mathbf{N}$                                                                        | $x \in \mathbf{N}$                                                          | mentation are laws of composition (the compositions of $x \in \mathbf{N}$ and $y \in \mathbf{N}$                                                           |
| kohlhase-omdoc_F00061 | 49   | ■ 1.000 | $y \in \mathbf{N}$                                                                        | $y \in \mathbf{N}$                                                          | mentation are laws of composition (the compositions of $x \in \mathbf{N}$ and $y \in \mathbf{N}$                                                           |
| kohlhase-omdoc_F00062 | 49   | ■ 0.997 | $x+y, x \ y$                                                                              | $x + y, xy$                                                                 | under these laws being denoted respectively by $x + y, xy$ , or $x.y$ and $x^y$ ) ( <i>Set</i>                                                             |
| kohlhase-omdoc_F00063 | 49   | ■ 0.997 | $x \cdot y$                                                                               | $x.y$                                                                       | under these laws being denoted respectively by $x + y, xy$ , or $x.y$ and $x^y$ ) ( <i>Set</i>                                                             |
| kohlhase-omdoc_F00064 | 49   | ■ 0.997 | $x^{\{y\}}$                                                                               | $x^y$                                                                       | under these laws being denoted respectively by $x + y, xy$ , or $x.y$ and $x^y$ ) ( <i>Set</i>                                                             |
| kohlhase-omdoc_F00065 | 49   | ■ 1.000 | $(X, Y) \mapsto X \circ Y$                                                                | $(X, Y) \mapsto X \circ Y$                                                  | (3) Let $E$ be a set; the mapping <u><math>(X, Y) \mapsto X \circ Y</math></u> is a law of composition                                                     |
| kohlhase-omdoc_F00066 | 49   | ■ 1.000 | $(f, g) \mapsto f \circ g$                                                                | $(f, g) \mapsto f \circ g$                                                  | mapping <u><math>(f, g) \mapsto f \circ g</math></u> is a law of composition on the set of mappings from $E$                                               |
| kohlhase-omdoc_F00067 | 52   | ■ 0.221 | $\mathrm{APT}_{\mathrm{E}}\mathrm{X}$                                                     | $\mathrm{APT}_{\mathrm{E}}\mathrm{X}$                                       | The text of the L <sup>A</sup> T <sub>E</sub> X fragment we are migrating. For simplic-                                                                    |
| kohlhase-omdoc_F00068 | 59   | ■ 0.999 | $\lambda$                                                                                 | $\lambda$                                                                   | a $\lambda$ -term $\lambda E, F. \mathit{set}(E) \wedge F : E \times E \rightarrow E^2$ . The value <b>simple</b> of the <b>type</b> at-                   |
| kohlhase-omdoc_F00069 | 59   | ■ 0.999 | $\lambda E, F. \mathit{operatorname{set}}(E) \wedge F : E \times E \rightarrow E^2$       | $\lambda E, F. \mathit{set}(E) \wedge F : E \times E \rightarrow E^2$       | a $\lambda$ -term <u><math>\lambda E, F. \mathit{set}(E) \wedge F : E \times E \rightarrow E^2</math></u> . The value <b>simple</b> of the <b>type</b> at- |
| kohlhase-omdoc_F00070 | 59   | ■ 1.000 | $(A, B)$                                                                                  | $(A, B)$                                                                    | it occurs. So if we have <code>law_of_composition(A, B)</code> somewhere this can be                                                                       |
| kohlhase-omdoc_F00071 | 59   | ■ 0.863 | $(\lambda E, F. \mathit{operatorname{set}}(E) \wedge F : E \times E \rightarrow E)(A, B)$ | $(\lambda E, F. \mathit{set}(E) \wedge F : E \times E \rightarrow E)(A, B)$ | reduced to <u><math>(\lambda E, F. \mathit{set}(E) \wedge F : E \times E \rightarrow E)(A, B)</math></u> which in turn reduces <sup>3</sup>                |
| kohlhase-omdoc_F00072 | 59   | ■ 0.917 | $\mathit{operatorname{set}}(A) \wedge B : A \times A \rightarrow A$                       | $\mathit{set}(A) \wedge B : A \times A \rightarrow A$                       | to <u><math>\mathit{set}(A) \wedge B : A \times A \rightarrow A</math></u> or in other words <code>law_of_composition(A, B)</code> is                      |
| kohlhase-omdoc_F00073 | 59   | ■ 1.000 | $A$                                                                                       | $A$                                                                         | true, iff $A$ is a set and $B$ is a function from $A \times A$ to $A$ . This definition is                                                                 |
| kohlhase-omdoc_F00074 | 59   | ■ 1.000 | $B$                                                                                       | $B$                                                                         | true, iff $A$ is a set and $B$ is a function from $A \times A$ to $A$ . This definition is                                                                 |
| kohlhase-omdoc_F00075 | 59   | ■ 1.000 | $A \times A$                                                                              | $A \times A$                                                                | true, iff $A$ is a set and $B$ is a function from $A \times A$ to $A$ . This definition is                                                                 |
| kohlhase-omdoc_F00076 | 59   | ■ 0.819 | $\lambda X. A$                                                                            | $\lambda X. A$                                                              | the form $\lambda X. A$ to an argument $B$ , then the result is obtained by binding all the                                                                |
| kohlhase-omdoc_F00077 | 59   | ■ 1.000 | $X$                                                                                       | $X$                                                                         | formal parameters $X$ to the actual parameter $B$ , i.e. the result is the value of $A$ ,                                                                  |
| kohlhase-omdoc_F00078 | 60   | ■ 0.967 | $M$                                                                                       | $M$                                                                         | in other words: $M$ is a magma, iff it is a pair $(E, F)$ , where $F$ is a law of                                                                          |
| kohlhase-omdoc_F00079 | 60   | ■ 0.967 | $(E, F)$                                                                                  | $(E, F)$                                                                    | in other words: $M$ is a magma, iff it is a pair $(E, F)$ , where $F$ is a law of                                                                          |

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| kohlhase-omdoc_F00080 | 60   | ■ 0.967 | F                                                                                   | $F$                                                                                 | in other words: $M$ is a magma, iff it is a pair $(E, F)$ , where $F$ is a law of                                                                           |
| kohlhase-omdoc_F00081 | 62   | ■ 0.986 | $\left[\mathrm{BCC}^{+04}\right]$                                                   | $\left[\mathrm{BCC}^{+04}\right]$                                                   | dard $\left[\mathrm{BCC}^{+04}\right]$ to codify knowledge about mathematical symbols and con-                                                              |
| kohlhase-omdoc_F00082 | 66   | ■ 1.000 | n                                                                                   | $n$                                                                                 | The symbol to represent the $n$ -ary function to return the least common mul-                                                                               |
| kohlhase-omdoc_F00083 | 66   | ■ 0.981 | $\operatorname{lcm}(a, b)$                                                          | $\operatorname{lcm}(a, b)$                                                          | We define $\operatorname{lcm}(a, b)$ as $\frac{a \cdot b}{\operatorname{gcd}(a, b)}$                                                                        |
| kohlhase-omdoc_F00084 | 66   | ■ 0.981 | $\frac{a \cdot b}{\operatorname{gcd}(a, b)}$                                        | $\frac{a \cdot b}{\operatorname{gcd}(a, b)}$                                        | We define $\operatorname{lcm}(a, b)$ as $\frac{a \cdot b}{\operatorname{gcd}(a, b)}$                                                                        |
| kohlhase-omdoc_F00085 | 66   | ■ 0.940 | a                                                                                   | $a$                                                                                 | <b>Lemma 3.</b> For all integers $a, b$ there is no $c > 0$ such that $(a c)$ and $(b c)$ and                                                               |
| kohlhase-omdoc_F00086 | 66   | ■ 0.940 | b                                                                                   | $b$                                                                                 | <b>Lemma 3.</b> For all integers $a, b$ there is no $c > 0$ such that $(a c)$ and $(b c)$ and                                                               |
| kohlhase-omdoc_F00087 | 66   | ■ 0.940 | $c > 0$                                                                             | $c > 0$                                                                             | <b>Lemma 3.</b> For all integers $a, b$ there is no $c > 0$ such that $(a c)$ and $(b c)$ and                                                               |
| kohlhase-omdoc_F00088 | 66   | ■ 0.940 | $(a \mid c)$                                                                        | $(a \mid c)$                                                                        | <b>Lemma 3.</b> For all integers $a, b$ there is no $c > 0$ such that $(a c)$ and $(b c)$ and                                                               |
| kohlhase-omdoc_F00089 | 66   | ■ 0.940 | $(b \mid c)$                                                                        | $(b \mid c)$                                                                        | <b>Lemma 3.</b> For all integers $a, b$ there is no $c > 0$ such that $(a c)$ and $(b c)$ and                                                               |
| kohlhase-omdoc_F00090 | 66   | ■ 1.000 | $c < \operatorname{lcm}(a, b)$                                                      | $c < \operatorname{lcm}(a, b)$                                                      | $c < \operatorname{lcm}(a, b)$ .                                                                                                                            |
| kohlhase-omdoc_F00091 | 66   | ■ 0.998 | $\operatorname{kgV}(a, b)$                                                          | $\operatorname{kgV}(a, b)$                                                          | Wir definieren $\operatorname{kgV}(a, b)$ als $\frac{a \cdot b}{\operatorname{ggT}(a, b)}$                                                                  |
| kohlhase-omdoc_F00092 | 66   | ■ 0.998 | $\frac{a \cdot b}{\operatorname{ggT}(a, b)}$                                        | $\frac{a \cdot b}{\operatorname{ggT}(a, b)}$                                        | Wir definieren $\operatorname{kgV}(a, b)$ als $\frac{a \cdot b}{\operatorname{ggT}(a, b)}$                                                                  |
| kohlhase-omdoc_F00093 | 66   | ■ 0.354 | $b \mid c$                                                                          | $b \mid c$                                                                          | <b>Lemma 3.</b> Für alle ganzen Zahlen $a, b$ gibt es kein $c > 0$ mit $(a c)$ und $(b c)$                                                                  |
| kohlhase-omdoc_F00094 | 66   | ■ 0.531 | $c < \operatorname{kgV}(a, b)$                                                      | $c < \operatorname{kgV}(a, b)$                                                      | und $c < \operatorname{kgV}(a, b)$ .                                                                                                                        |
| kohlhase-omdoc_F00095 | 68   | ■ 1.000 | $\mathbb{N}$                                                                        | $\mathbb{N}$                                                                        | more elaborate definition of $\mathbb{N}$ ). The <code>sortdef</code> element posits that the set of                                                        |
| kohlhase-omdoc_F00097 | 68   | ■ 1.000 | s                                                                                   | $s$                                                                                 | instance the number three is represented as $s(s(s(0)))$ , where $s$ stands for                                                                             |
| kohlhase-omdoc_F00098 | 68   | ■ 1.000 | $\leq$                                                                              | $\leq$                                                                              | Finally we have extended the natural numbers by an ordering function $\leq$                                                                                 |
| kohlhase-omdoc_F00099 | 69   | ■ 0.887 | $\mathrm{T0}^1$                                                                     | $\mathrm{T0}^1$                                                                     | theory, as witnessed by the assertion $\operatorname{leq}.\mathrm{T0}^1$ .                                                                                  |
| kohlhase-omdoc_F00100 | 72   | ■ 1.000 | $\sigma(\circ)$                                                                     | $\sigma(\circ)$                                                                     | together with the converse $\sigma(\circ)$ of $\circ$ is also a semigroup: the obligation for                                                               |
| kohlhase-omdoc_F00101 | 72   | ■ 1.000 | $\sigma(\forall x, y \in M. x \circ y \in M) = \forall y, x \in M. x \circ y \in M$ | $\sigma(\forall x, y \in M. x \circ y \in M) = \forall y, x \in M. x \circ y \in M$ | $\sigma(\forall x, y \in M. x \circ y \in M) = \forall y, x \in M. x \circ y \in M$ , which is logically equivalent                                         |
| kohlhase-omdoc_F00102 | 72   | ■ 0.835 | $\tau x$                                                                            | $\tau x$                                                                            | scription operator $\tau x$ .: If there is a unique $x$ , such that <b>A</b> is true, then the                                                              |
| kohlhase-omdoc_F00103 | 73   | ■ 1.000 | $\rho$                                                                              | $\rho$                                                                              | $\rho$ in Figure 7.1), in OMDoc we can share this structure using the <code>base on</code>                                                                  |
| kohlhase-omdoc_F00104 | 73   | ■ 0.951 | $\cdot^{-1}$                                                                        | $\cdot^{-1}$                                                                        | function $\cdot^{-1}$ from that via a definite description.                                                                                                 |
| kohlhase-omdoc_F00105 | 74   | ■ 1.000 | $\sigma$                                                                            | $\sigma$                                                                            | represented by $\sigma$ and $\tau$ in Figure 7.1). These rename all of the imported                                                                         |
| kohlhase-omdoc_F00106 | 74   | ■ 1.000 | $\tau$                                                                              | $\tau$                                                                              | represented by $\sigma$ and $\tau$ in Figure 7.1). These rename all of the imported                                                                         |
| kohlhase-omdoc_F00107 | 74   | ■ 0.301 | $\cdot^*$                                                                           | $\cdot^*$                                                                           | the punctuated set constructor $\cdot^*$ , which is imported from the additive group                                                                        |
| kohlhase-omdoc_F00108 | 74   | ■ 0.267 | $l$                                                                                 | $l$                                                                                 | $\langle \operatorname{morphism} \rangle M \mapsto R, x \circ y \mapsto x * y, e \mapsto 1, \cdot^{-1} \mapsto - \langle / \operatorname{morphism} \rangle$ |

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|-----------------------|------|---------|----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| kohlhase-omdoc_F00109 | 84   | ■ 1.000 | $\mathcal{N}$                                                                                                                    | $\mathcal{N}$                                                                                                                    | imation $\mathcal{N}$ of the real theory $\mathcal{F}$ , only to show an example $\mathcal{E}_{\mathcal{N}}$ of where this                                                           |
| kohlhase-omdoc_F00110 | 84   | ■ 1.000 | $\mathcal{F}$                                                                                                                    | $\mathcal{F}$                                                                                                                    | imation $\mathcal{N}$ of the real theory $\mathcal{F}$ , only to show an example $\mathcal{E}_{\mathcal{N}}$ of where this                                                           |
| kohlhase-omdoc_F00111 | 84   | ■ 1.000 | $\mathcal{E}_{\mathcal{N}}$                                                                                                      | $\mathcal{E}_{\mathcal{N}}$                                                                                                      | imation $\mathcal{N}$ of the real theory $\mathcal{F}$ , only to show an example $\mathcal{E}_{\mathcal{N}}$ of where this                                                           |
| kohlhase-omdoc_F00112 | 84   | ■ 1.000 | $\mathcal{S}$                                                                                                                    | $\mathcal{S}$                                                                                                                    | is insufficient. Then we propose a first (straw-man) solution $\mathcal{S}$ , and show an                                                                                            |
| kohlhase-omdoc_F00113 | 84   | ■ 1.000 | $\mathcal{E}_{\mathcal{S}}$                                                                                                      | $\mathcal{E}_{\mathcal{S}}$                                                                                                      | example $\mathcal{E}_{\mathcal{S}}$ of why this does not work. Based on the information we gleaned                                                                                   |
| kohlhase-omdoc_F00114 | 84   | ■ 1.000 | $\mathcal{E}_{\mathcal{F}}$                                                                                                      | $\mathcal{E}_{\mathcal{F}}$                                                                                                      | theory and demonstrate that this works on $\mathcal{E}_{\mathcal{F}}$ .                                                                                                              |
| kohlhase-omdoc_F00115 | 84   | ■ 1.000 | $\mathcal{N}, \mathcal{E}_{\mathcal{N}}, \mathcal{S}, \mathcal{E}_{\mathcal{S}}, \mathcal{F}$                                    | $\mathcal{N}, \mathcal{E}_{\mathcal{N}}, \mathcal{S}, \mathcal{E}_{\mathcal{S}}, \mathcal{F}$                                    | sents the content structure, where the boxes $\mathcal{N}, \mathcal{E}_{\mathcal{N}}, \mathcal{S}, \mathcal{E}_{\mathcal{S}}, \mathcal{F}$ , and $\mathcal{E}_{\mathcal{F}}$ signify |
| kohlhase-omdoc_F00116 | 84   | ■ 0.384 | $\operatorname{slides} \mathrm{sl}_i$                                                                                            | slides $\mathrm{sl}_i$                                                                                                           | the slides $\mathrm{sl}_i$ are grouped into a lecture. The dashed lines between the two                                                                                              |
| kohlhase-omdoc_F00117 | 84   | ■ 1.000 | $\mathrm{n}_1$                                                                                                                   | $\mathrm{n}_1$                                                                                                                   | example: the text fragment $\mathrm{n}_1$ links the content $\mathcal{E}_{\mathcal{N}}$ , which is referenced from                                                                   |
| kohlhase-omdoc_F00118 | 84   | ■ 0.974 | $\mathrm{n}_2$                                                                                                                   | $\mathrm{n}_2$                                                                                                                   | the content structure, to slide 1. The fragment $\mathrm{n}_2$ might say something like                                                                                              |
| kohlhase-omdoc_F00119 | 85   | ■ 0.986 | $\left[\operatorname{MBG}^{+03}\right]$                                                                                          | $\left[\operatorname{MBG}^{+03}\right]$                                                                                          | the ACTIVEMATH system $\left[\operatorname{MBG}^{+03}\right]$ generates a simple narrative structure                                                                                 |
| kohlhase-omdoc_F00120 | 87   | ■ 0.810 | $9.3^2$                                                                                                                          | $9.3^2$                                                                                                                          | in Listing 9.3 <sup>2</sup> to the URL which uniquely identifies the service object in the                                                                                           |
| kohlhase-omdoc_F00121 | 103  | ■ 1.000 | $R$                                                                                                                              | $R$                                                                                                                              | Let $R$ be a ref element of type <code>include</code> . We call the element the URI                                                                                                  |
| kohlhase-omdoc_F00122 | 103  | ■ 0.978 | $\left[\operatorname{SBC}^{+00}\right]$                                                                                          | $\left[\operatorname{SBC}^{+00}\right]$                                                                                          | system $\left[\operatorname{SBC}^{+00}\right]$ can — for instance — extract document fragments based on                                                                              |
| kohlhase-omdoc_F00123 | 107  | ■ 0.910 | $\langle YYY Y \rangle - \langle MM \rangle - \langle DD \rangle T \langle hh \rangle : \langle mm \rangle : \langle ss \rangle$ | $\langle YYY Y \rangle - \langle MM \rangle - \langle DD \rangle T \langle hh \rangle : \langle mm \rangle : \langle ss \rangle$ | Concretely, the format is $\langle YYY Y \rangle - \langle MM \rangle - \langle DD \rangle T \langle hh \rangle : \langle mm \rangle : \langle ss \rangle$                           |
| kohlhase-omdoc_F00124 | 107  | ■ 0.781 | $\langle YYY Y \rangle$                                                                                                          | $\langle YYY Y \rangle$                                                                                                          | where $\langle YYY Y \rangle$ represents the year, $\langle MM \rangle$ the month, and $\langle DD \rangle$ the                                                                      |
| kohlhase-omdoc_F00125 | 107  | ■ 0.781 | $\langle MM \rangle$                                                                                                             | $\langle MM \rangle$                                                                                                             | where $\langle YYY Y \rangle$ represents the year, $\langle MM \rangle$ the month, and $\langle DD \rangle$ the                                                                      |
| kohlhase-omdoc_F00126 | 107  | ■ 0.781 | $\langle DD \rangle$                                                                                                             | $\langle DD \rangle$                                                                                                             | where $\langle YYY Y \rangle$ represents the year, $\langle MM \rangle$ the month, and $\langle DD \rangle$ the                                                                      |
| kohlhase-omdoc_F00127 | 107  | ■ 0.582 | $\langle hh \rangle, \langle mm \rangle, \langle ss \rangle$                                                                     | $\langle hh \rangle, \langle mm \rangle, \langle ss \rangle$                                                                     | date/time separator and $\langle hh \rangle, \langle mm \rangle, \langle ss \rangle$ represent hour, minutes, and                                                                    |
| kohlhase-omdoc_F00128 | 107  | ■ 0.260 | $\langle ss \rangle . \langle sss \rangle$                                                                                       | $\langle ss \rangle . \langle sss \rangle$                                                                                       | cision of fractional seconds if desired, i.e the format $\langle ss \rangle . \langle sss \rangle$ with                                                                              |
| kohlhase-omdoc_F00129 | 108  | ■ 1.000 | $\widehat{=}$                                                                                                                    | $\widehat{=}$                                                                                                                    | be an ISO 639 norm two-letter language specifier, like <code>en</code> $\widehat{=}$ English,                                                                                        |
| kohlhase-omdoc_F00130 | 108  | ■ 0.327 | $\mathrm{de} \widehat{=}$                                                                                                        | $\mathrm{de} \widehat{=}$                                                                                                        | <code>de</code> $\widehat{=}$ German, <code>fr</code> $\widehat{=}$ French, <code>nl</code> $\widehat{=}$ Dutch, . . .                                                               |
| kohlhase-omdoc_F00131 | 114  | ■ 0.586 | $^{+03}$                                                                                                                         | $^{+03}$                                                                                                                         | and the MATHML 2.0 recommendation (second edition) $\left[\operatorname{ABC}^{+03}\right]$ are part                                                                                  |
| kohlhase-omdoc_F00132 | 116  | ■ 1.000 | $\sin(x)$                                                                                                                        | $\sin(x)$                                                                                                                        | For instance, the formula $\sin(x)$ would be modeled as an application of                                                                                                            |
| kohlhase-omdoc_F00133 | 116  | ■ 1.000 | $\forall x. \sin(x) \leq \pi$                                                                                                    | $\forall x. \sin(x) \leq \pi$                                                                                                    | formula $\forall x. \sin(x) \leq \pi$ .                                                                                                                                              |
| kohlhase-omdoc_F00134 | 117  | ■ 0.999 | $x + \pi$                                                                                                                        | $x + \pi$                                                                                                                        | In the OPENMATH fragment in Listing 13.3 the expression $x + \pi$ is annotated                                                                                                       |

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| kohlhase-omdoc_F00135 | 118  | ■ 1.000 | $\sin (\pi)$                                                                          | $\sin(\pi)$                                                                           | faulty representation of $\sin(\pi)$ . The outer error flags an arity violation (the                                                                                                                      |
| kohlhase-omdoc_F00136 | 118  | ■ 1.000 | $\pi$                                                                                 | $\pi$                                                                                 | the representation of the constant $\pi$ (we used the name <b>po</b> instead of <b>pi</b> ).                                                                                                              |
| kohlhase-omdoc_F00137 | 121  | ■ 0.973 | $14^{\text{th}}$                                                                      | 14 <sup>th</sup>                                                                      | metics (Kindergarten to 14 <sup>th</sup> grade (i.e. undergraduate college level)).                                                                                                                       |
| kohlhase-omdoc_F00138 | 121  | ■ 0.630 | $2.0\left[\mathrm{ABC}^{+}03\right]$                                                  | 2.0 [ABC <sup>+</sup> 03]                                                             | second edition of MATHML 2.0 [ABC <sup>+</sup> 03], a maintenance release for the                                                                                                                         |
| kohlhase-omdoc_F00139 | 121  | ■ 0.997 | $\left[\mathrm{ABC}^{+}03\right]$                                                     | [ABC <sup>+</sup> 03]                                                                 | MATHML 2.0 that are relevant to OMDoc; for the full story see <span style="border: 1px solid red;">[ABC<sup>+</sup>03]</span> .                                                                           |
| kohlhase-omdoc_F00140 | 121  | ■ 0.456 | $^{\{7\}}$                                                                            | $\overline{7}$                                                                        | The top-level element of MATHML is the <b>m:math</b> <sup>7</sup> element, see Figure 13.10                                                                                                               |
| kohlhase-omdoc_F00141 | 124  | ■ 1.000 | $(\mathbb{R} \rightarrow \mathbb{R}) \rightarrow (\mathbb{R} \rightarrow \mathbb{R})$ | $(\mathbb{R} \rightarrow \mathbb{R}) \rightarrow (\mathbb{R} \rightarrow \mathbb{R})$ | e.g. the type <span style="border: 1px solid red;"><math>(\mathbb{R} \rightarrow \mathbb{R}) \rightarrow (\mathbb{R} \rightarrow \mathbb{R})</math></span> for functionals on the real numbers. In        |
| kohlhase-omdoc_F00142 | 126  | ■ 0.397 | $\mathrm{cd} =$                                                                       | cd =                                                                                  | tent dictionary identified in the cd attribute. A symbol $\langle \mathrm{OMS} \mathrm{cd} = \langle cd_1 \rangle$                                                                                        |
| kohlhase-omdoc_F00143 | 126  | ■ 0.397 | $\left\langle \left\langle cd_1 \right\rangle \right\rangle$                          | $\langle \langle cd_1 \rangle \rangle$                                                | tent dictionary identified in the cd attribute. A symbol $\langle \mathrm{OMS} \mathrm{cd} = \langle cd_1 \rangle$                                                                                        |
| kohlhase-omdoc_F00144 | 126  | ■ 0.175 | $\left\langle \left\langle cd_2 \right\rangle \right\rangle$                          | $\langle \langle cd_2 \rangle \rangle$                                                | $\mathrm{name} = \langle \langle name_1 \rangle \rangle />$ is equal to $\langle \mathrm{OMS} \mathrm{cd} = \langle cd_2 \rangle$ $\mathrm{name} = \langle \langle name_2 \rangle \rangle />$ , iff       |
| kohlhase-omdoc_F00145 | 126  | ■ 0.175 | $\left\langle \left\langle name_2 \right\rangle \right\rangle$                        | $\langle \langle name_2 \rangle \rangle$                                              | $\mathrm{name} = \langle \langle name_1 \rangle \rangle />$ is equal to $\langle \mathrm{OMS} \mathrm{cd} = \langle cd_2 \rangle$ $\mathrm{name} = \langle \langle name_2 \rangle \rangle />$ , iff       |
| kohlhase-omdoc_F00146 | 126  | ■ 0.973 | $\langle \langle cd_1 \rangle \rangle = \langle \langle cd_2 \rangle \rangle$         | $\langle \langle cd_1 \rangle \rangle = \langle \langle cd_2 \rangle \rangle$         | $\langle \langle cd_1 \rangle \rangle = \langle \langle cd_2 \rangle \rangle$ and $\langle \langle name_1 \rangle \rangle = \langle \langle name_2 \rangle \rangle$ as XML simple names. In case of vari- |
| kohlhase-omdoc_F00147 | 126  | ■ 0.973 | $\langle \langle name_1 \rangle \rangle = \langle \langle name_2 \rangle \rangle$     | $\langle \langle name_1 \rangle \rangle = \langle \langle name_2 \rangle \rangle$     | $\langle \langle cd_1 \rangle \rangle = \langle \langle cd_2 \rangle \rangle$ and $\langle \langle name_1 \rangle \rangle = \langle \langle name_2 \rangle \rangle$ as XML simple names. In case of vari- |
| kohlhase-omdoc_F00148 | 126  | ■ 0.923 | $\alpha$                                                                              | $\alpha$                                                                              | named without changing the object ( <b><math>\alpha</math>-conversion</b> ). If $\langle \mathrm{OMV} \mathrm{name} = \mathrm{x} \rangle />$                                                              |
| kohlhase-omdoc_F00149 | 129  | ■ 0.990 | $\mathrm{CMP}^2$                                                                      | $\mathrm{CMP}^2$                                                                      | is represented as a groups of <b>CMP</b> <sup>2</sup> elements which contain the vernacular                                                                                                               |
| kohlhase-omdoc_F00150 | 130  | ■ 0.942 | $\mathrm{FMP}^4$                                                                      | $\mathrm{FMP}^4$                                                                      | An <b>FMP</b> <sup>4</sup> element is the general element for representing formal mathematical                                                                                                            |
| kohlhase-omdoc_F00151 | 131  | ■ 0.287 | $V, F. \mathrm{Im}(F) = V$                                                            | $V, F. \mathrm{Im}(F) = V$                                                            | $\langle \mathrm{FMP} \mathrm{logic} = \text{“hol”} \rangle \text{binop} = \lambda V, F. \mathrm{Im}(F) = V \wedge \mathrm{Dom}(F) = V \langle / \mathrm{FMP} \rangle$                                    |
| kohlhase-omdoc_F00152 | 131  | ■ 0.404 | $A \langle / \mathrm{FMP} \rangle$                                                    | $A \langle / \mathrm{FMP} \rangle$                                                    | is equivalent to $\langle \mathrm{FMP} \rangle A \langle / \mathrm{FMP} \rangle$ , where $A$ is an OPENMATH, Content-MATHML,                                                                              |
| kohlhase-omdoc_F00153 | 134  | ■ 0.562 | $0 \in \mathbb{N}$                                                                    | $0 \in \mathbb{N}$                                                                    | ken <sup>5</sup> . Let us consider an example: We can equivalently say “ $0 \in \mathbb{N}$ ” and “the                                                                                                    |
| kohlhase-omdoc_F00154 | 137  | ■ 0.995 | $\underline{\mathrm{ul}}$                                                             | $\underline{\mathrm{ul}}$                                                             | <i>Unordered Lists</i> <b>ul</b> is the constructor for unordered or bulleted lists, the in                                                                                                               |
| kohlhase-omdoc_F00155 | 140  | ■ 1.000 | $s(n)$                                                                                | $s(n)$                                                                                | 2. The successor $s(n)$ of a natural number $n$ is a natural number.                                                                                                                                      |
| kohlhase-omdoc_F00156 | 140  | ■ 0.997 | $0, s$                                                                                | $0, s$                                                                                | structures that interpret 0, $s$ , and $\mathbb{N}$ and satisfy these axioms must be isomor-                                                                                                              |
| kohlhase-omdoc_F00157 | 141  | ■ 1.000 | $\mathcal{T}$                                                                         | $\mathcal{T}$                                                                         | of a theory $\mathcal{T}$ by just looking at its form: for instance, axioms of the form $s =$                                                                                                             |
| kohlhase-omdoc_F00158 | 141  | ■ 1.000 | $s =$                                                                                 | $s =$                                                                                 | of a theory $\mathcal{T}$ by just looking at its form: for instance, axioms of the form $s =$                                                                                                             |
| kohlhase-omdoc_F00159 | 141  | ■ 0.623 | $\mathbf{A}$                                                                          | <b>A</b>                                                                              | <b>A</b> , where $s$ is a symbol that does not occur in $\mathcal{T}$ and <b>A</b> is a formula containing                                                                                                |
| kohlhase-omdoc_F00160 | 141  | ■ 1.000 | $s = \mathbf{A}$                                                                      | $s = \mathbf{A}$                                                                      | symbols. The axiom $s = \mathbf{A}$ only constrains the meaning of the <b>new symbol</b>                                                                                                                  |
| kohlhase-omdoc_F00161 | 141  | ■ 1.000 | $\mathcal{A}$                                                                         | $\mathcal{A}$                                                                         | Generally an axiom $\mathcal{A}$ that results in a conservative extension is called                                                                                                                       |

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| kohlhase-omdoc_F00163 | 141  | ■ 1.000 | $e^{\cdot}$                                                                               | $e^{\cdot}$                                                                               | $e^{\cdot}$                                                                                                                                                                                  |
| kohlhase-omdoc_F00164 | 141  | ■ 0.554 | $\partial f = f[$                                                                         | $\partial f = f[$                                                                         | the differential equation $\partial f = f$ [where $f(0) = 1$ ].                                                                                                                              |
| kohlhase-omdoc_F00165 | 141  | ■ 0.554 | $f(0)=1]$                                                                                 | $f(0) = 1]$                                                                               | the differential equation $\partial f = f$ [where $f(0) = 1$ ].                                                                                                                              |
| kohlhase-omdoc_F00166 | 141  | ■ 1.000 | $x+\theta=x$                                                                              | $x + 0 = x$                                                                               | the equations $x + 0 = x$ and $x + s(y) = s(x + y)$ .                                                                                                                                        |
| kohlhase-omdoc_F00167 | 141  | ■ 1.000 | $x+s(y)=s(x+y)$                                                                           | $x + s(y) = s(x + y)$                                                                     | the equations $x + 0 = x$ and $x + s(y) = s(x + y)$ .                                                                                                                                        |
| kohlhase-omdoc_F00168 | 143  | ■ 0.993 | $1.1\left[\mathrm{BPSM}^{+}04\right]$                                                     | 1.1 [BPSM <sup>+</sup> 04]                                                                | [1.1 BPSM <sup>+</sup> 04]. The optional attribute <b>scope</b> takes the values <b>global</b> and                                                                                           |
| kohlhase-omdoc_F00169 | 144  | ■ 1.000 | $e: t$                                                                                    | $e : t$                                                                                   | processes. A <b>type declaration</b> $e: t$ makes the information that a symbol or                                                                                                           |
| kohlhase-omdoc_F00170 | 145  | ■ 0.998 | $t$                                                                                       | $t$                                                                                       | expression $e$ is in a set represented by a type $t$ available to a specified math-                                                                                                          |
| kohlhase-omdoc_F00171 | 145  | ■ 1.000 | $\sum_{i=1}^n a_i x^i$                                                                    | $\sum_{i=1}^n a_i x^i$                                                                    | or that expressions of the form $\sum_{i=1}^n a_i x^i$ are polynomials, if the $a_i$ are real                                                                                                |
| kohlhase-omdoc_F00172 | 145  | ■ 1.000 | $a_i$                                                                                     | $a_i$                                                                                     | or that expressions of the form $\sum_{i=1}^n a_i x^i$ are polynomials, if the $a_i$ are real                                                                                                |
| kohlhase-omdoc_F00173 | 145  | ■ 0.999 | $\forall x_1, \dots, x_n. e \in t$                                                        | $\forall x_1, \dots, x_n. e \in t$                                                        | $\forall x_1, \dots, x_n. e \in t$ for a type declaration $e: t$ with variables $x_1, \dots, x_n$ . If the                                                                                   |
| kohlhase-omdoc_F00174 | 145  | ■ 0.999 | $x_1, \dots, x_n$                                                                         | $x_1, \dots, x_n$                                                                         | $\forall x_1, \dots, x_n. e \in t$ for a type declaration $e: t$ with variables $x_1, \dots, x_n$ . If the                                                                                   |
| kohlhase-omdoc_F00176 | 149  | ■ 0.944 | $x+x$                                                                                     | $x + x$                                                                                   | Listing 15.14 contains a type assertion $x+x: evens$ , which makes the infor-                                                                                                                |
| kohlhase-omdoc_F00177 | 150  | ■ 1.000 | $\mathcal{C}_i$                                                                           | $\mathcal{C}_i$                                                                           | and $\mathcal{C} := \{\mathcal{C}_1, \dots, \mathcal{C}_m\}$ , and $\mathcal{F}^{-1}$ be the sequent that has the $\mathcal{C}_i$ as assump-                                                 |
| kohlhase-omdoc_F00180 | 150  | ■ 1.000 | $\mathcal{F}^{-1}$                                                                        | $\mathcal{F}^{-1}$                                                                        | and $\mathcal{C} := \{\mathcal{C}_1, \dots, \mathcal{C}_m\}$ , and $\mathcal{F}^{-1}$ be the sequent that has the $\mathcal{C}_i$ as assump-                                                 |
| kohlhase-omdoc_F00181 | 308  | ■ 1.000 | $\mathcal{C}$                                                                             | $\mathcal{C}$                                                                             | (i.e. a set of well-formed formulae) and a calculus $\mathcal{C}$ (i.e. a set of inference                                                                                                   |
| kohlhase-omdoc_F00186 | 150  | ■ 0.999 | $\mathcal{A}_1, \dots, \mathcal{A}_n$                                                     | $\mathcal{A}_1, \dots, \mathcal{A}_n$                                                     | Where $\mathcal{F}$ is an assertion whose FMP has <b>assumption</b> elements $\mathcal{A}_1, \dots, \mathcal{A}_n$                                                                           |
| kohlhase-omdoc_F00187 | 150  | ■ 0.993 | $\mathcal{C}_1, \dots, \mathcal{C}_m$                                                     | $\mathcal{C}_1, \dots, \mathcal{C}_m$                                                     | and <b>conclusion</b> elements $\mathcal{C}_1, \dots, \mathcal{C}_m$ . Furthermore, let $\mathcal{A} := \{\mathcal{A}_1, \dots, \mathcal{A}_n\}$                                             |
| kohlhase-omdoc_F00188 | 150  | ■ 0.993 | $\mathcal{A} := \{\mathcal{A}_1, \dots, \mathcal{A}_n\}$                                  | $\mathcal{A} := \{\mathcal{A}_1, \dots, \mathcal{A}_n\}$                                  | and <b>conclusion</b> elements $\mathcal{C}_1, \dots, \mathcal{C}_m$ . Furthermore, let $\mathcal{A} := \{\mathcal{A}_1, \dots, \mathcal{A}_n\}$                                             |
| kohlhase-omdoc_F00189 | 150  | ■ 1.000 | $\mathcal{C} := \{\mathcal{C}_1, \dots, \mathcal{C}_m\}$                                  | $\mathcal{C} := \{\mathcal{C}_1, \dots, \mathcal{C}_m\}$                                  | and $\mathcal{C} := \{\mathcal{C}_1, \dots, \mathcal{C}_m\}$ , and $\mathcal{F}^{-1}$ be the sequent that has the $\mathcal{C}_i$ as assump-                                                 |
| kohlhase-omdoc_F00190 | 150  | ■ 1.000 | $\mathcal{A}_i$                                                                           | $\mathcal{A}_i$                                                                           | tions and the $\mathcal{A}_i$ as conclusions. Finally, let $\overline{\mathcal{C}} := \{\overline{\mathcal{C}}_1, \dots, \overline{\mathcal{C}}_m\}$ , where $\overline{\mathcal{C}}_i$ is a |
| kohlhase-omdoc_F00191 | 150  | ■ 1.000 | $\overline{\mathcal{C}} := \{\overline{\mathcal{C}}_1, \dots, \overline{\mathcal{C}}_m\}$ | $\overline{\mathcal{C}} := \{\overline{\mathcal{C}}_1, \dots, \overline{\mathcal{C}}_m\}$ | tions and the $\mathcal{A}_i$ as conclusions. Finally, let $\overline{\mathcal{C}} := \{\overline{\mathcal{C}}_1, \dots, \overline{\mathcal{C}}_m\}$ , where $\overline{\mathcal{C}}_i$ is a |
| kohlhase-omdoc_F00192 | 150  | ■ 1.000 | $\overline{\mathcal{C}}_i$                                                                | $\overline{\mathcal{C}}_i$                                                                | tions and the $\mathcal{A}_i$ as conclusions. Finally, let $\overline{\mathcal{C}} := \{\overline{\mathcal{C}}_1, \dots, \overline{\mathcal{C}}_m\}$ , where $\overline{\mathcal{C}}_i$ is a |
| kohlhase-omdoc_F00193 | 152  | ■ 0.997 | $\mathcal{E}$                                                                             | $\mathcal{E}$                                                                             | objects in OMDoc. Conceptually, we model an example $\mathcal{E}$ as a pair $(\mathcal{W}, \mathbf{A})$ ,                                                                                    |
| kohlhase-omdoc_F00194 | 152  | ■ 0.997 | $(\mathcal{W}, \mathbf{A})$                                                               | $(\mathcal{W}, \mathbf{A})$                                                               | objects in OMDoc. Conceptually, we model an example $\mathcal{E}$ as a pair $(\mathcal{W}, \mathbf{A})$ ,                                                                                    |
| kohlhase-omdoc_F00195 | 152  | ■ 0.992 | $\mathcal{W} = (\mathcal{W}_1, \dots, \mathcal{W}_n)$                                     | $\mathcal{W} = (\mathcal{W}_1, \dots, \mathcal{W}_n)$                                     | where $\mathcal{W} = (\mathcal{W}_1, \dots, \mathcal{W}_n)$ is an $n$ -tuple of mathematical objects and $\mathbf{A}$ is an                                                                  |
| kohlhase-omdoc_F00196 | 152  | ■ 0.923 | $\mathbf{S}$                                                                              | $\mathbf{S}$                                                                              | symbol $\mathbf{S}$ , then $\mathbf{A}$ must be of the form $\mathbf{S}(\mathcal{W}_1, \dots, \mathcal{W}_n)$ .                                                                              |

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| kohlhase-omdoc_F00197 | 152  | ■ 0.923 | $\mathbf{S}\left(\mathcal{W}_{\{1\}}, \ldots, \mathcal{W}_{\{n\}}\right)$                  | $\mathbf{S}(\mathcal{W}_1, \dots, \mathcal{W}_n)$               | symbol $\mathbf{S}$ , then $\mathbf{A}$ must be of the form $\mathbf{S}(\mathcal{W}_1, \dots, \mathcal{W}_n)$ .                                            |
| kohlhase-omdoc_F00198 | 152  | ■ 0.999 | $\mathbf{C}$                                                                               | $\mathbf{C}$                                                    | tion more carefully. We assume that $\mathbf{C}$ is of the form $\mathcal{Q}\mathbf{D}$ for some formula $\mathbf{D}$ ,                                    |
| kohlhase-omdoc_F00199 | 152  | ■ 0.999 | $\mathcal{Q} \ \mathbf{D}$                                                                 | $\mathcal{Q}\mathbf{D}$                                         | tion more carefully. We assume that $\mathbf{C}$ is of the form $\mathcal{Q}\mathbf{D}$ for some formula $\mathbf{D}$ ,                                    |
| kohlhase-omdoc_F00200 | 152  | ■ 0.999 | $\mathbf{D}$                                                                               | $\mathbf{D}$                                                    | tion more carefully. We assume that $\mathbf{C}$ is of the form $\mathcal{Q}\mathbf{D}$ for some formula $\mathbf{D}$ ,                                    |
| kohlhase-omdoc_F00201 | 152  | ■ 1.000 | $\mathcal{Q}$                                                                              | $\mathcal{Q}$                                                   | where $\mathcal{Q}$ is a sequence $\mathcal{Q}_1\mathcal{W}_1, \dots, \mathcal{Q}_m\mathcal{W}_m$ of $m \geq n = \#\mathcal{W}$ quantifications of         |
| kohlhase-omdoc_F00202 | 152  | ■ 1.000 | $\mathcal{Q}_{\{1\}} \mathcal{W}_{\{1\}}, \ldots, \mathcal{Q}_{\{m\}} \mathcal{W}_{\{m\}}$ | $\mathcal{Q}_1\mathcal{W}_1, \dots, \mathcal{Q}_m\mathcal{W}_m$ | where $\mathcal{Q}$ is a sequence $\mathcal{Q}_1\mathcal{W}_1, \dots, \mathcal{Q}_m\mathcal{W}_m$ of $m \geq n = \#\mathcal{W}$ quantifications of         |
| kohlhase-omdoc_F00203 | 152  | ■ 1.000 | $m \geq n = \#\mathcal{W}$                                                                 | $m \geq n = \#\mathcal{W}$                                      | where $\mathcal{Q}$ is a sequence $\mathcal{Q}_1\mathcal{W}_1, \dots, \mathcal{Q}_m\mathcal{W}_m$ of $m \geq n = \#\mathcal{W}$ quantifications of         |
| kohlhase-omdoc_F00204 | 152  | ■ 0.999 | $\mathcal{Q}_{\{i\}}$                                                                      | $\mathcal{Q}_i$                                                 | using quantifiers $\mathcal{Q}_i$ like $\forall$ or $\exists$ . Now let $\mathcal{Q}'$ be a sub-sequence of $m - n$ quan-                                  |
| kohlhase-omdoc_F00205 | 152  | ■ 0.999 | $\forall$                                                                                  | $\forall$                                                       | using quantifiers $\mathcal{Q}_i$ like $\forall$ or $\exists$ . Now let $\mathcal{Q}'$ be a sub-sequence of $m - n$ quan-                                  |
| kohlhase-omdoc_F00206 | 152  | ■ 0.999 | $\exists$                                                                                  | $\exists$                                                       | using quantifiers $\mathcal{Q}_i$ like $\forall$ or $\exists$ . Now let $\mathcal{Q}'$ be a sub-sequence of $m - n$ quan-                                  |
| kohlhase-omdoc_F00207 | 152  | ■ 0.999 | $\mathcal{Q}^{\prime}$                                                                     | $\mathcal{Q}'$                                                  | using quantifiers $\mathcal{Q}_i$ like $\forall$ or $\exists$ . Now let $\mathcal{Q}'$ be a sub-sequence of $m - n$ quan-                                  |
| kohlhase-omdoc_F00208 | 152  | ■ 0.999 | $m - n$                                                                                    | $m - n$                                                         | using quantifiers $\mathcal{Q}_i$ like $\forall$ or $\exists$ . Now let $\mathcal{Q}'$ be a sub-sequence of $m - n$ quan-                                  |
| kohlhase-omdoc_F00209 | 152  | ■ 0.985 | $\mathbf{D}^{\prime}$                                                                      | $\mathbf{D}'$                                                   | tifiers of $\mathcal{Q}$ and $\mathbf{D}'$ be $\mathbf{D}$ only that all the $\mathcal{W}_{i_j}$ such that the $\mathcal{Q}_{i_j}$ are absent              |
| kohlhase-omdoc_F00210 | 152  | ■ 0.985 | $\mathcal{W}_{\{i_j\}}$                                                                    | $\mathcal{W}_{i_j}$                                             | tifiers of $\mathcal{Q}$ and $\mathbf{D}'$ be $\mathbf{D}$ only that all the $\mathcal{W}_{i_j}$ such that the $\mathcal{Q}_{i_j}$ are absent              |
| kohlhase-omdoc_F00211 | 152  | ■ 0.985 | $\mathcal{Q}_{\{i_j\}}$                                                                    | $\mathcal{Q}_{i_j}$                                             | tifiers of $\mathcal{Q}$ and $\mathbf{D}'$ be $\mathbf{D}$ only that all the $\mathcal{W}_{i_j}$ such that the $\mathcal{Q}_{i_j}$ are absent              |
| kohlhase-omdoc_F00212 | 152  | ■ 0.998 | $\mathcal{W}_{\{j\}}$                                                                      | $\mathcal{W}_j$                                                 | from $\mathcal{Q}'$ have been replaced by $\mathcal{W}_j$ for $1 \leq j \leq n$ . If $\mathcal{E} = (\mathcal{W}, \mathbf{A})$ supports $\mathbf{C}$ ,     |
| kohlhase-omdoc_F00213 | 152  | ■ 0.998 | $1 \leq j \leq n$                                                                          | $1 \leq j \leq n$                                               | from $\mathcal{Q}'$ have been replaced by $\mathcal{W}_j$ for $1 \leq j \leq n$ . If $\mathcal{E} = (\mathcal{W}, \mathbf{A})$ supports $\mathbf{C}$ ,     |
| kohlhase-omdoc_F00214 | 152  | ■ 0.998 | $\mathcal{E} = (\mathcal{W}, \mathbf{A})$                                                  | $\mathcal{E} = (\mathcal{W}, \mathbf{A})$                       | from $\mathcal{Q}'$ have been replaced by $\mathcal{W}_j$ for $1 \leq j \leq n$ . If $\mathcal{E} = (\mathcal{W}, \mathbf{A})$ supports $\mathbf{C}$ ,     |
| kohlhase-omdoc_F00215 | 152  | ■ 1.000 | $\mathbf{A} = \mathcal{Q}^{\prime} \mathbf{D}^{\prime}$                                    | $\mathbf{A} = \mathcal{Q}'\mathbf{D}'$                          | then $\mathbf{A} = \mathcal{Q}'\mathbf{D}'$ and if $\mathcal{E}$ is a counter-example for $\mathbf{C}$ , then $\mathbf{A} = \neg\mathcal{Q}'\mathbf{D}'$ . |
| kohlhase-omdoc_F00216 | 152  | ■ 1.000 | $\mathbf{A} = \neg \mathcal{Q}^{\prime} \mathbf{D}^{\prime}$                               | $\mathbf{A} = \neg\mathcal{Q}'\mathbf{D}'$                      | then $\mathbf{A} = \mathcal{Q}'\mathbf{D}'$ and if $\mathcal{E}$ is a counter-example for $\mathbf{C}$ , then $\mathbf{A} = \neg\mathcal{Q}'\mathbf{D}'$ . |
| kohlhase-omdoc_F00217 | 154  | ■ 1.000 | $A^*$                                                                                      | $A^*$                                                           | alphabet $A$ as an argument and returns the set $A^*$ of stringss over $A$ , <code>concat</code>                                                           |
| kohlhase-omdoc_F00218 | 154  | ■ 0.914 | $\epsilon$                                                                                 | $\epsilon$                                                      | the empty string $\epsilon$ . Then we state that $\mathcal{W} = (A^*, ::, \epsilon)$ is a monoid in an                                                     |
| kohlhase-omdoc_F00219 | 154  | ■ 0.914 | $\mathcal{W} = \left(A^*, ::, \epsilon\right)$                                             | $\mathcal{W} = (A^*, ::, \epsilon)$                             | the empty string $\epsilon$ . Then we state that $\mathcal{W} = (A^*, ::, \epsilon)$ is a monoid in an                                                     |
| kohlhase-omdoc_F00220 | 154  | ■ 1.000 | $\left(\mathcal{W}, \mathbf{A}^{\prime}\right)$                                            | $(\mathcal{W}, \mathbf{A}')$                                    | <code>mon.ex2</code> uses the pair $(\mathcal{W}, \mathbf{A}')$ as a counter-example to the false conjecture                                               |
| kohlhase-omdoc_F00221 | 154  | ■ 0.380 | $M,$                                                                                       | $M,$                                                            | <CMP>In a monoid $M$ , the left unit and the right unit coincide,</CMP>                                                                                    |
| kohlhase-omdoc_F00223 | 161  | ■ 1.000 | $\emptyset$                                                                                | $\emptyset$                                                     | tors $\emptyset$ and the set insertion function $\iota$ , since the set $\{a\}$ can be obtained                                                            |
| kohlhase-omdoc_F00224 | 161  | ■ 1.000 | $\iota$                                                                                    | $\iota$                                                         | tors $\emptyset$ and the set insertion function $\iota$ , since the set $\{a\}$ can be obtained                                                            |
| kohlhase-omdoc_F00225 | 161  | ■ 1.000 | $\{a\}$                                                                                    | $\{a\}$                                                         | tors $\emptyset$ and the set insertion function $\iota$ , since the set $\{a\}$ can be obtained                                                            |

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|-----------------------|------|---------|-------------------------------------------------------|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| kohlhase-omdoc_F00226 | 161  | ■ 1.000 | $\iota(a, \emptyset) = \iota(a, \iota(a, \emptyset))$ | $\iota(a, \emptyset) = \iota(a, \iota(a, \emptyset))$ | e.g. $\iota(a, \emptyset) = \iota(a, \iota(a, \emptyset))$ . This kind of sort is called <b>generated</b> , since it only                        |
| kohlhase-omdoc_F00227 | 162  | ■ 1.000 | $\mathbb{P}$                                          | $\mathbb{P}$                                          | encoding. In this example, we introduce a second sort $\mathbb{P}$ for positive nat-                                                             |
| kohlhase-omdoc_F00228 | 166  | ■ 1.000 | $P$                                                   | $P$                                                   | <b>Proof:</b> We need to prove that the set $P$ of all prime numbers is not                                                                      |
| kohlhase-omdoc_F00229 | 166  | ■ 1.000 | $P = \{p_1, \dots, p_n\}$                             | $P = \{p_1, \dots, p_n\}$                             | 3. Then $P = \{p_1, \dots, p_n\}$ for some $p_i$ .                                                                                               |
| kohlhase-omdoc_F00230 | 166  | ■ 1.000 | $p_i$                                                 | $p_i$                                                 | 3. Then $P = \{p_1, \dots, p_n\}$ for some $p_i$ .                                                                                               |
| kohlhase-omdoc_F00231 | 166  | ■ 1.000 | $q \stackrel{\text{def}}{=} p_1 \cdots p_n + 1$       | $q \stackrel{\text{def}}{=} p_1 \cdots p_n + 1$       | 4. Let $q \stackrel{\text{def}}{=} p_1 \cdots p_n + 1$ .                                                                                         |
| kohlhase-omdoc_F00232 | 166  | ■ 1.000 | $p_i \in P$                                           | $p_i \in P$                                           | 5. Since for each $p_i \in P$ we have $q > p_i$ , we conclude $q \notin P$ .                                                                     |
| kohlhase-omdoc_F00233 | 166  | ■ 1.000 | $q > p_i$                                             | $q > p_i$                                             | 5. Since for each $p_i \in P$ we have $q > p_i$ , we conclude $q \notin P$ .                                                                     |
| kohlhase-omdoc_F00234 | 166  | ■ 1.000 | $q \notin P$                                          | $q \notin P$                                          | 5. Since for each $p_i \in P$ we have $q > p_i$ , we conclude $q \notin P$ .                                                                     |
| kohlhase-omdoc_F00235 | 166  | ■ 1.000 | $q$                                                   | $q$                                                   | 6. We prove the absurdity by showing that $q$ is prime:                                                                                          |
| kohlhase-omdoc_F00236 | 166  | ■ 1.000 | $q = p_i k + 1$                                       | $q = p_i k + 1$                                       | 7. For each $p_i \in P$ we have $q = p_i k + 1$ for some natural                                                                                 |
| kohlhase-omdoc_F00237 | 166  | ■ 1.000 | $k$                                                   | $k$                                                   | number $k$ , so $p_i$ can not divide $q$ ;                                                                                                       |
| kohlhase-omdoc_F00238 | 166  | ■ 0.990 | $\bar{\lambda}\mu\tilde{\mu}$                         | $\bar{\lambda}\mu\tilde{\mu}$                         | by interpreting it as an implementation of the $\bar{\lambda}\mu\tilde{\mu}$ calculus [Coe05] proof                                              |
| kohlhase-omdoc_F00239 | 168  | ■ 1.000 | $12ff$                                                | $12ff$                                                | the proof. See for example lines $12ff$ in Listing 17.2 for examples of                                                                          |
| kohlhase-omdoc_F00240 | 170  | ■ 0.986 | $[\text{BCF}^+97]$                                    | $[\text{BCF}^+97]$                                    | multiple levels of abstraction). This proof representation (see [BCF <sup>+</sup> 97] for a                                                      |
| kohlhase-omdoc_F00241 | 179  | ■ 0.791 | $(R, +)$                                              | $(R, +)$                                              | $R$ together with two operations $+$ and $*$ , such that $(R, +)$ is a group with                                                                |
| kohlhase-omdoc_F00242 | 179  | ■ 1.000 | $(R^*, *)$                                            | $(R^*, *)$                                            | unit 0 and inverse operation $-$ and $(R^*, *)$ is a monoid with unit 1 and base                                                                 |
| kohlhase-omdoc_F00243 | 179  | ■ 1.000 | $R^* := \{r \in R \mid r \neq 0\}$                    | $R^* := \{r \in R \mid r \neq 0\}$                    | set $R^* := \{r \in R \mid r \neq 0\}$ . Using the literal inheritance regime introduced so                                                      |
| kohlhase-omdoc_F00244 | 179  | ■ 0.604 | $M = (S, \circ, e)$                                   | $M = (S, \circ, e)$                                   | <sup>2</sup> On any monoid $M = (S, \circ, e)$ , we have the $\cdot^*$ operator, which converts a set                                            |
| kohlhase-omdoc_F00245 | 179  | ■ 1.000 | $S \subseteq M$                                       | $S \subseteq M$                                       | $S \subseteq M$ in to $S^* := \{r \in S \mid r \neq e\}$                                                                                         |
| kohlhase-omdoc_F00246 | 179  | ■ 1.000 | $S^* := \{r \in S \mid r \neq e\}$                    | $S^* := \{r \in S \mid r \neq e\}$                    | $S \subseteq M$ in to $S^* := \{r \in S \mid r \neq e\}$                                                                                         |
| kohlhase-omdoc_F00247 | 180  | ■ 1.000 | $\{t_1, \dots, t_n\}$                                 | ${}^3t_1, \dots, t_n$                                 | Let $\mathcal{S}$ be a theory with theory-constitutive elements <sup>3</sup> $t_1, \dots, t_n$ and $\sigma: \mathcal{S} \rightarrow \mathcal{T}$ |
| kohlhase-omdoc_F00248 | 180  | ■ 1.000 | $\sigma: \mathcal{S} \rightarrow \mathcal{T}$         | $\sigma: \mathcal{S} \rightarrow \mathcal{T}$         | Let $\mathcal{S}$ be a theory with theory-constitutive elements <sup>3</sup> $t_1, \dots, t_n$ and $\sigma: \mathcal{S} \rightarrow \mathcal{T}$ |
| kohlhase-omdoc_F00249 | 180  | ■ 1.000 | $\sigma(t_i)$                                         | $\sigma(t_i)$                                         | constitutive statements $\sigma(t_i)$ from $\mathcal{S}$ . For instance, the theory of rings inherits                                            |
| kohlhase-omdoc_F00250 | 180  | ■ 1.000 | $\forall x. x + 0 = x$                                | $\forall x. x + 0 = x$                                | the axiom $\forall x. x + 0 = x$ from the theory of monoids as $\sigma(\forall x. x + 0 = x) =$                                                  |
| kohlhase-omdoc_F00251 | 180  | ■ 1.000 | $\sigma(\forall x. x + 0 = x) =$                      | $\sigma(\forall x. x + 0 = x) =$                      | the axiom $\forall x. x + 0 = x$ from the theory of monoids as $\sigma(\forall x. x + 0 = x) =$                                                  |
| kohlhase-omdoc_F00252 | 180  | ■ 0.999 | $\forall x. x * 1 = x$                                | $\forall x. x * 1 = x$                                | $\forall x. x * 1 = x$ .                                                                                                                         |
| kohlhase-omdoc_F00253 | 181  | ■ 0.999 | $R, +, 0, -, *, 1$                                    | $R, +, 0, -, *, 1$                                    | the morphism. In our example, the symbols $R, +, 0, -, *, 1$ are visible in the                                                                  |

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|-----------------------|------|---------|--------------------------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| kohlhase-omdoc_F00254 | 181  | ■ 1.000 | L                                                                  | $L$                                                                | concept of a sorting function, i.e. a function that — given a list $L$ as input                                                                   |
| kohlhase-omdoc_F00255 | 181  | ■ 1.000 | $L^{\{\prime\}}$                                                   | $L'$                                                               | — returns a returns a permutation $L'$ of $L$ that is ordered. In the situation                                                                   |
| kohlhase-omdoc_F00256 | 182  | ■ 0.879 | $\sigma_{\{i\}}$                                                   | $\sigma_i$                                                         | $\sigma_i$ with <code>xml:id="«refi»"</code> , such that the target theory of $\sigma_{i-1}$ is the source                                        |
| kohlhase-omdoc_F00257 | 182  | ■ 0.879 | $\sigma_{\{i-1\}}$                                                 | $\sigma_{i-1}$                                                     | $\sigma_i$ with <code>xml:id="«refi»"</code> , such that the target theory of $\sigma_{i-1}$ is the source                                        |
| kohlhase-omdoc_F00258 | 182  | ■ 0.902 | $\mathcal{S}, \sigma(\mathbf{S})$                                  | $\mathcal{S}, \sigma(\mathbf{S})$                                  | constitutive statement $\mathbf{S}$ of $\mathcal{S}$ , $\sigma(\mathbf{S})$ is provable in $\mathcal{T}$ (we say that $\sigma(\mathbf{S})$ is a   |
| kohlhase-omdoc_F00259 | 182  | ■ 0.902 | $\sigma(\mathbf{S})$                                               | $\sigma(\mathbf{S})$                                               | constitutive statement $\mathbf{S}$ of $\mathcal{S}$ , $\sigma(\mathbf{S})$ is provable in $\mathcal{T}$ (we say that $\sigma(\mathbf{S})$ is a   |
| kohlhase-omdoc_F00260 | 182  | ■ 0.905 | $\mathbf{T}$                                                       | $\mathbf{T}$                                                       | in theory $\mathcal{T}$ via $\sigma$ , if there is an assertional statement $\mathbf{T}$ in $\mathcal{T}$ , such that the                         |
| kohlhase-omdoc_F00261 | 182  | ■ 1.000 | $\mathcal{T}^5$                                                    | $\mathcal{T}^5$                                                    | discussed in Section 18.1 induces a theory inclusion from $\mathcal{S}$ into $\mathcal{T}^5$ (the                                                 |
| kohlhase-omdoc_F00262 | 183  | ■ 1.000 | $\sigma(\mathbf{A})$                                               | $\sigma(\mathbf{A})$                                               | $\mathcal{S}$ -theorem for for all $\mathcal{T}$ -theorems of the from $\sigma(\mathbf{A})$ . If the morphism $\sigma$ is the                     |
| kohlhase-omdoc_F00263 | 183  | ■ 0.421 | $\forall x, y, z \in M. z \circ (y \circ x) = (z \circ y) \circ x$ | $\forall x, y, z \in M. z \circ (y \circ x) = (z \circ y) \circ x$ | <code>&lt;assertion xml:id="conv.assoc"&gt;<math>\forall x, y, z \in M. z \circ (y \circ x) = (z \circ y) \circ x</math>&lt;/assertion&gt;</code> |
| kohlhase-omdoc_F00264 | 183  | ■ 0.421 | $\forall x, y, z \in M. z \circ (y \circ x) = (z \circ y) \circ x$ | $\forall x, y, z \in M. z \circ (y \circ x) = (z \circ y) \circ x$ | <code>&lt;assertion xml:id="conv.assoc"&gt;<math>\forall x, y, z \in M. z \circ (y \circ x) = (z \circ y) \circ x</math>&lt;/assertion&gt;</code> |
| kohlhase-omdoc_F00265 | 183  | ■ 0.765 | $\forall x, y \in M. y \circ x \in M$                              | $\forall x, y \in M. y \circ x \in M$                              | <code>&lt;assertion xml:id="conv.closed" theory="semigroup"&gt;<math>\forall x, y \in M. y \circ x \in M</math>&lt;/assertion&gt;</code>          |
| kohlhase-omdoc_F00266 | 183  | ■ 0.325 | $\forall x \in M. e \circ x = x$                                   | $\forall x \in M. e \circ x = x$                                   | <code>&lt;assertion xml:id="left.unit" theory="monoid"&gt;<math>\forall x \in M. e \circ x = x</math>&lt;/assertion&gt;</code>                    |
| kohlhase-omdoc_F00267 | 186  | ■ 0.998 | $\mathrm{BCD}^{+02}$                                               | $\mathrm{BCD}^{+02}$                                               | (see e.g. $\mathrm{BCD}^{+02}$ for an overview over the branch cut situation). Reading                                                            |
| kohlhase-omdoc_F00268 | 187  | ■ 1.000 | $\Gamma$                                                           | $\Gamma$                                                           | in the top theory inclusion $\Gamma$ . On the basis of theories $\mathcal{T}_1$ and $\mathcal{T}_2$ , theory $\mathcal{C}_1$                      |
| kohlhase-omdoc_F00269 | 187  | ■ 1.000 | $\mathcal{T}_1$                                                    | $\mathcal{T}_1$                                                    | in the top theory inclusion $\Gamma$ . On the basis of theories $\mathcal{T}_1$ and $\mathcal{T}_2$ , theory $\mathcal{C}_1$                      |
| kohlhase-omdoc_F00270 | 187  | ■ 1.000 | $\mathcal{T}_2$                                                    | $\mathcal{T}_2$                                                    | in the top theory inclusion $\Gamma$ . On the basis of theories $\mathcal{T}_1$ and $\mathcal{T}_2$ , theory $\mathcal{C}_1$                      |
| kohlhase-omdoc_F00271 | 187  | ■ 1.000 | $\mathcal{C}_1$                                                    | $\mathcal{C}_1$                                                    | in the top theory inclusion $\Gamma$ . On the basis of theories $\mathcal{T}_1$ and $\mathcal{T}_2$ , theory $\mathcal{C}_1$                      |
| kohlhase-omdoc_F00272 | 187  | ■ 0.999 | $\mathcal{A}_1$                                                    | $\mathcal{A}_1$                                                    | is built up via theories $\mathcal{A}_1$ and $\mathcal{B}_1$ . Similarly, theory $\mathcal{C}_2$ is built up via $\mathcal{A}_2$                  |
| kohlhase-omdoc_F00273 | 187  | ■ 0.999 | $\mathcal{B}_1$                                                    | $\mathcal{B}_1$                                                    | is built up via theories $\mathcal{A}_1$ and $\mathcal{B}_1$ . Similarly, theory $\mathcal{C}_2$ is built up via $\mathcal{A}_2$                  |
| kohlhase-omdoc_F00274 | 187  | ■ 0.999 | $\mathcal{C}_2$                                                    | $\mathcal{C}_2$                                                    | is built up via theories $\mathcal{A}_1$ and $\mathcal{B}_1$ . Similarly, theory $\mathcal{C}_2$ is built up via $\mathcal{A}_2$                  |
| kohlhase-omdoc_F00275 | 187  | ■ 0.999 | $\mathcal{A}_2$                                                    | $\mathcal{A}_2$                                                    | is built up via theories $\mathcal{A}_1$ and $\mathcal{B}_1$ . Similarly, theory $\mathcal{C}_2$ is built up via $\mathcal{A}_2$                  |
| kohlhase-omdoc_F00276 | 187  | ■ 1.000 | $\mathcal{B}_2$                                                    | $\mathcal{B}_2$                                                    | and $\mathcal{B}_2$ (in the latter, we have a non-trivial non-trivial morphism $\sigma$ ). Let us                                                 |
| kohlhase-omdoc_F00277 | 187  | ■ 1.000 | $\mathcal{X}_i \in \{\mathcal{A}, \mathcal{B}, \mathcal{C}\}$      | $\mathcal{X}_i \in \{\mathcal{A}, \mathcal{B}, \mathcal{C}\}$      | assume for the sake of this argument that for $\mathcal{X}_i \in \{\mathcal{A}, \mathcal{B}, \mathcal{C}\}$ theories $\mathcal{X}_1$ and          |
| kohlhase-omdoc_F00278 | 187  | ■ 1.000 | $\mathcal{X}_1$                                                    | $\mathcal{X}_1$                                                    | assume for the sake of this argument that for $\mathcal{X}_i \in \{\mathcal{A}, \mathcal{B}, \mathcal{C}\}$ theories $\mathcal{X}_1$ and          |
| kohlhase-omdoc_F00279 | 187  | ■ 1.000 | $\mathcal{X}_2$                                                    | $\mathcal{X}_2$                                                    | $\mathcal{X}_2$ are so similar that axiom inclusions (they are indicated by thin dashed                                                           |
| kohlhase-omdoc_F00280 | 187  | ■ 0.995 | $\alpha, \beta$                                                    | $\alpha, \beta$                                                    | arrows in Figure 18.8 and have the formula-mappings $\alpha$ , $\beta$ , and $\gamma$ ) are easy                                                  |
| kohlhase-omdoc_F00281 | 187  | ■ 0.995 | $\gamma$                                                           | $\gamma$                                                           | arrows in Figure 18.8 and have the formula-mappings $\alpha$ , $\beta$ , and $\gamma$ ) are easy                                                  |

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| kohlhase-omdoc_<br>F00282 | 187  | ■ 1.000 | <code>\beta(\mathbf{B})</code>                                                              | $\beta(\mathbf{B})$                                          | constitutive for $\mathcal{C}_1$ , say that it is local in $\mathcal{B}_1$ , then we already know that $\beta(\mathbf{B})$ is                               |
| kohlhase-omdoc_<br>F00283 | 187  | ■ 1.000 | <code>\beta</code>                                                                          | $\beta$                                                      | provable in $\mathcal{B}_2$ since $\beta$ is an axiom inclusion. Moreover, we know that $\sigma(\beta(\mathbf{B}))$                                         |
| kohlhase-omdoc_<br>F00284 | 187  | ■ 1.000 | <code>\sigma(\beta(\mathbf{B}))</code>                                                      | $\sigma(\beta(\mathbf{B}))$                                  | provable in $\mathcal{B}_2$ since $\beta$ is an axiom inclusion. Moreover, we know that $\sigma(\beta(\mathbf{B}))$                                         |
| kohlhase-omdoc_<br>F00285 | 187  | ■ 1.000 | <code>\Gamma=\sigma \circ \beta</code>                                                      | $\Gamma = \sigma \circ \beta$                                | have $\Gamma = \sigma \circ \beta$ , then we are done for $\mathbf{B}$ and in fact for all local statements                                                 |
| kohlhase-omdoc_<br>F00286 | 187  | ■ 1.000 | <code>\Gamma^{-1}</code>                                                                    | $\Gamma^{-1}$                                                | (and often also in the converse theory inclusion $\Gamma^{-1}$ that would prove that the                                                                    |
| kohlhase-omdoc_<br>F00287 | 188  | ■ 1.000 | <code>\tau:=\sigma_n \circ \cdots \circ \sigma_1 \circ \sigma</code><br><code>\sigma</code> | $\tau := \sigma_n \circ \cdots \circ \sigma_1 \circ \sigma$  | morphism $\tau := \sigma_n \circ \cdots \circ \sigma_1 \circ \sigma$ ), if $\mathcal{S} \xrightarrow{\sigma} \mathcal{T}_1$ is an axiom inclusion or (local |
| kohlhase-omdoc_<br>F00288 | 188  | ■ 1.000 | <code>\mathcal{S} \xrightarrow{\sigma} \mathcal{T}_1</code>                                 | $\mathcal{S} \xrightarrow{\sigma} \mathcal{T}_1$             | morphism $\tau := \sigma_n \circ \cdots \circ \sigma_1 \circ \sigma$ ), if $\mathcal{S} \xrightarrow{\sigma} \mathcal{T}_1$ is an axiom inclusion or (local |
| kohlhase-omdoc_<br>F00289 | 188  | ■ 1.000 | <code>\mathcal{T}_i \xrightarrow{\sigma_i} \mathcal{T}_{i+1}</code>                         | $\mathcal{T}_i \xrightarrow{\sigma_i} \mathcal{T}_{i+1}$     | theory import) and $\mathcal{T}_i \xrightarrow{\sigma_i} \mathcal{T}_{i+1}$ are theory inclusions (or local theory import).                                 |
| kohlhase-omdoc_<br>F00290 | 188  | ■ 1.000 | <code>\mathcal{T}_1, \mathcal{T}_2, \mathcal{A}_1</code>                                    | $\mathcal{T}_1, \mathcal{T}_2, \mathcal{A}_1$                | justify four new axiom inclusions from the theories $\mathcal{T}_1$ , $\mathcal{T}_2$ , $\mathcal{A}_1$ , and $\mathcal{B}_1$ into                          |
| kohlhase-omdoc_<br>F00291 | 188  | ■ 1.000 | <code>\mathcal{X}</code>                                                                    | $\mathcal{X}$                                                | Thus, for each theory $\mathcal{X}$ that $\mathcal{C}_1$ inherits from, there is an axiom inclusion                                                         |
| kohlhase-omdoc_<br>F00292 | 188  | ■ 1.000 | <code>\mathcal{S} \xrightarrow{\sigma} \mathcal{T}</code>                                   | $\mathcal{S} \xrightarrow{\sigma} \mathcal{T}$               | We call a situation where we have a formula mapping $\mathcal{S} \xrightarrow{\sigma} \mathcal{T}$ , and                                                    |
| kohlhase-omdoc_<br>F00293 | 188  | ■ 1.000 | <code>\mathcal{X} \xrightarrow{\sigma_{\mathcal{X}}} \mathcal{T}</code>                     | $\mathcal{X} \xrightarrow{\sigma_{\mathcal{X}}} \mathcal{T}$ | an axiom inclusion $\mathcal{X} \xrightarrow{\sigma_{\mathcal{X}}} \mathcal{T}$ for every theory $\mathcal{X}$ that $\mathcal{S}$ inherits from a           |
| kohlhase-omdoc_<br>F00294 | 188  | ■ 1.000 | <code>\sigma_{\mathcal{X}}</code>                                                           | $\sigma_{\mathcal{X}}$                                       | <b>decomposition</b> for $\sigma$ , if the $\sigma_{\mathcal{X}}$ and $\sigma$ are compatible. As we have seen in                                           |
| kohlhase-omdoc_<br>F00295 | 190  | ■ 0.996 | <code>i</code>                                                                              | $i$                                                          | there must be theory inclusions $\sigma_i$ with <code>xml:id="refi"</code> , such that the target                                                           |
| kohlhase-omdoc_<br>F00296 | 190  | ■ 1.000 | <code>\sigma_1</code>                                                                       | $\sigma_1$                                                   | theory of $\sigma_1$ is the source theory of $\sigma_2$ .                                                                                                   |
| kohlhase-omdoc_<br>F00297 | 190  | ■ 1.000 | <code>\sigma_2</code>                                                                       | $\sigma_2$                                                   | theory of $\sigma_1$ is the source theory of $\sigma_2$ .                                                                                                   |
| kohlhase-omdoc_<br>F00298 | 197  | ■ 1.000 | <code>\forall P, Q. P \vee Q \Rightarrow Q \vee P</code>                                    | $\forall P, Q. P \vee Q \Rightarrow Q \vee P$                | <del><math>\forall P, Q. P \vee Q \Rightarrow Q \vee P</math></del> assuming appropriate templates for implication and                                      |
| kohlhase-omdoc_<br>F00299 | 199  | ■ 1.000 | <code>g c d</code>                                                                          | $gcd$                                                        | $gcd$ in English but $ggT$ (“größter gemeinsamer Teiler”) in German). Obvi-                                                                                 |
| kohlhase-omdoc_<br>F00300 | 199  | ■ 1.000 | <code>g g T</code>                                                                          | $ggT$                                                        | $gcd$ in English but $ggT$ (“größter gemeinsamer Teiler”) in German). Obvi-                                                                                 |
| kohlhase-omdoc_<br>F00301 | 200  | ■ 1.000 | <code>f^{\prime}</code>                                                                     | $f'$                                                         | $f'$ . The case <code>infix</code> is reserved for binary operators, where the function is                                                                  |
| kohlhase-omdoc_<br>F00306 | 201  | ■ 1.000 | <code>(a, b)</code>                                                                         | $(a, b)$                                                     | presentation. For instance, in the notation $(a, b)$ for pairs, the binary                                                                                  |
| kohlhase-omdoc_<br>F00307 | 201  | ■ 0.912 | <code>\mathrm{IAT}_{\mathrm{E}}\mathrm{X}</code>                                            | $\mathrm{IAT}_{\mathrm{E}}\mathrm{X}$                        | of the figure: the $\mathrm{IAT}_{\mathrm{E}}\mathrm{X}$ and the HTML presentations have attached                                                           |
| kohlhase-omdoc_<br>F00308 | 202  | ■ 0.369 | <code>\forall X</code>                                                                      | $\forall X$                                                  | MATH expression yields $\forall X.\text{true}$ , where the glyph $\forall$ carries a hyperlink <sup>6</sup> to                                              |
| kohlhase-omdoc_<br>F00309 | 202  | ■ 0.369 | <code>{ }^{\varnothing}</code>                                                              | $\varnothing$                                                | MATH expression yields $\forall X.\text{true}$ , where the glyph $\forall$ carries a hyperlink <sup>6</sup> to                                              |
| kohlhase-omdoc_<br>F00310 | 202  | ■ 1.000 | <code>\binom{n}{m}</code>                                                                   | $\binom{n}{m}$                                               | For instance, the binomial coefficient is some presented as $\binom{n}{m}$ (spoken                                                                          |
| kohlhase-omdoc_<br>F00311 | 202  | ■ 1.000 | <code>m</code>                                                                              | $m$                                                          | “ $n$ choose $m$ ”) and represented as                                                                                                                      |
| kohlhase-omdoc_<br>F00312 | 204  | ■ 0.995 | <code>X_{4}</code>                                                                          | $X_4$                                                        | tifier would be represented as $X_4$ , yielding $\forall X_4. X_4 = X_4$ which is exactly                                                                   |

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| kohlhase-omdoc_F00313 | 204  | ■ 0.995 | $\forall x_4. x_4 = x_4$                                                             | $\forall X_4. X_4 = X_4$                          | tifier would be represented as $X_4$ , yielding $\forall X_4. X_4 = X_4$ which is exactly                                                                                            |
| kohlhase-omdoc_F00314 | 205  | ■ 0.982 | $\forall \prec, X. X \prec X$                                                        | $\forall \prec, X. X \prec X$                     | This would give us the presentation $\forall \prec, X. X \prec X$ . Here, the first occurrence                                                                                       |
| kohlhase-omdoc_F00315 | 205  | ■ 1.000 | $\prec$                                                                              | $\prec$                                           | of the variable $\prec$ is handled by the second notation declaration (it does not                                                                                                   |
| kohlhase-omdoc_F00317 | 215  | ■ 0.891 | $\{\backslash, p \mid \hbox{$p$ and $p+2$ are prime}\backslash, \}$                  | <i>(not rendered)</i>                             | $\{\backslash, p \mid \hbox{$p$ and $p+2$ are prime}\backslash, \}$                                                                                                                  |
| kohlhase-omdoc_F00318 | 215  | ■ 0.891 | and                                                                                  | <i>and</i>                                        | $\{\backslash, p \mid \hbox{$p$ and $p+2$ are prime}\backslash, \}$                                                                                                                  |
| kohlhase-omdoc_F00319 | 215  | ■ 0.891 | are prime $\backslash, \}$                                                           | <i>(not rendered)</i>                             | $\{\backslash, p \mid \hbox{$p$ and $p+2$ are prime}\backslash, \}$                                                                                                                  |
| kohlhase-omdoc_F00320 | 215  | ■ 0.468 | $\{\backslash, p \mid p$                                                             | $\{p \mid p$                                      | $\{\backslash, p \mid p \text{~and } p+2 \text{ are prime} \backslash, \}$                                                                                                           |
| kohlhase-omdoc_F00321 | 215  | ■ 0.468 | $p+2$                                                                                | $p + 2$                                           | $\{\backslash, p \mid p \text{~and } p+2 \text{ are prime} \backslash, \}$                                                                                                           |
| kohlhase-omdoc_F00322 | 215  | ■ 0.468 | $\backslash, \}$                                                                     |                                                   | $\{\backslash, p \mid p \text{~and } p+2 \text{ are prime} \backslash, \}$                                                                                                           |
| kohlhase-omdoc_F00323 | 215  | ■ 0.405 | $\{\backslash, p \mid p \backslash \{\rm and\} p+2 \rm \ are \ prime \backslash, \}$ | <i>(not rendered)</i>                             | $\{\backslash, p \mid p \backslash \{\rm and\} \ p+2 \rm \ are \ prime \backslash, \}$ </phrase>                                                                                     |
| kohlhase-omdoc_F00324 | 218  | ■ 1.000 | $\mathcal{K}$                                                                        | $\mathcal{K}$                                     | For a formal definition let $\mathcal{K}$ be a set of documents. We take a <b>docu-</b>                                                                                              |
| kohlhase-omdoc_F00325 | 218  | ■ 0.935 | $\{ \}^{\{1\}} \mathcal{X}$                                                          | $^1 \mathcal{X}$                                  | <b>ment model</b> to be a partial equivalence relation <sup>1</sup> $\mathcal{X}$ on documents, such                                                                                 |
| kohlhase-omdoc_F00326 | 218  | ■ 0.998 | $\{d \mid d \in \mathcal{X}\} = \mathcal{K}$                                         | $\{d \mid d \in \mathcal{X}\} = \mathcal{K}$      | that $\{d \mid d \in \mathcal{X}\} = \mathcal{K}$ . In particular, a relation $\mathcal{X}$ is an equivalence relation on                                                            |
| kohlhase-omdoc_F00327 | 218  | ■ 1.000 | $d$                                                                                  | $d$                                               | $\mathcal{K}$ . For a given document model $\mathcal{X}$ , let us say that two documents $d$ and $d'$                                                                                |
| kohlhase-omdoc_F00328 | 218  | ■ 1.000 | $d^{\{\prime\}}$                                                                     | $d'$                                              | $\mathcal{K}$ . For a given document model $\mathcal{X}$ , let us say that two documents $d$ and $d'$                                                                                |
| kohlhase-omdoc_F00329 | 218  | ■ 1.000 | $d \in \mathcal{X}, d^{\{\prime\}}$                                                  | $d \mathcal{X} d'$                                | are <b><math>\mathcal{X}</math>-equal</b> , iff $d \mathcal{X} d'$ . We call a property $p$ <b><math>\mathcal{X}</math>-invariant</b> , iff for all $d \mathcal{X} d', p$            |
| kohlhase-omdoc_F00330 | 218  | ■ 1.000 | $p \in \mathcal{X}$                                                                  | $p \mathcal{X}$                                   | are <b><math>\mathcal{X}</math>-equal</b> , iff $d \mathcal{X} d'$ . We call a property $p$ <b><math>\mathcal{X}</math>-invariant</b> , iff for all $d \mathcal{X} d', p$            |
| kohlhase-omdoc_F00331 | 218  | ■ 1.000 | $d \in \mathcal{X}, d^{\{\prime\}}, p$                                               | $d \mathcal{X} d', p$                             | are <b><math>\mathcal{X}</math>-equal</b> , iff $d \mathcal{X} d'$ . We call a property $p$ <b><math>\mathcal{X}</math>-invariant</b> , iff for all $d \mathcal{X} d', p$            |
| kohlhase-omdoc_F00332 | 218  | ■ 1.000 | $p$                                                                                  | $p$                                               | holds on $d$ whenever $p$ holds on $d'$ .                                                                                                                                            |
| kohlhase-omdoc_F00333 | 218  | ■ 1.000 | $[d]_{\mathcal{X}}$                                                                  | $[d]_{\mathcal{X}}$                               | <sup>1</sup> A partial equivalence relation is a symmetric transitive relation. We will use $[d]_{\mathcal{X}}$                                                                      |
| kohlhase-omdoc_F00334 | 218  | ■ 0.999 | $[d]_{\mathcal{X}} := \{e \mid d \mathcal{X} e\}$                                    | $[d]_{\mathcal{X}} := \{e \mid d \mathcal{X} e\}$ | for the <b>equivalence class</b> of $d$ , i.e. $[d]_{\mathcal{X}} := \{e \mid d \mathcal{X} e\}$                                                                                     |
| kohlhase-omdoc_F00335 | 220  | ■ 1.000 | $X+Y=$                                                                               | $X + Y =$                                         | <i>not</i> use mathematical equality here, which would make the formula $X + Y =$                                                                                                    |
| kohlhase-omdoc_F00336 | 220  | ■ 0.268 | $Y+X$                                                                                | $Y + X$                                           | $Y + X$ (the <code>om:OMOBJ</code> with <code>xml:id="comm1"</code> in Listing 22.3 instantiated with                                                                                |
| kohlhase-omdoc_F00337 | 220  | ■ 1.000 | $X+Y=X+Y$                                                                            | $X + Y = X + Y$                                   | addition for <code>op</code> ) mathematically equal to the trivial condition $X+Y = X+Y$ ,                                                                                           |
| kohlhase-omdoc_F00338 | 220  | ■ 1.000 | $X+Y$                                                                                | $X + Y$                                           | obtained by exchanging the right hand side $Y + X$ of the equality by $X + Y$ ,                                                                                                      |
| kohlhase-omdoc_F00339 | 232  | ■ 0.999 | $\left[\mathrm{ABD}^{+01}\right]$                                                    | $[\mathrm{ABD}^{+01}]$                            | tions for DTD modularization $[\mathrm{ABD}^{+01}]$ . This partitions the DTD into <b>DTD</b>                                                                                        |
| kohlhase-omdoc_F00340 | 232  | ■ 0.966 | $\{ \}^{\{+\}} 01$                                                                   | $^{+01}$                                          | process, we will briefly describe the module structure. Following $[\mathrm{ABD}^{+01}]$ ,                                                                                           |
| kohlhase-omdoc_F00341 | 234  | ■ 0.988 | $d \rangle$                                                                          | $d \rangle$                                       | as <code>omdoc-<math>\langle \langle \mathit{SIId} \rangle \rangle</math>.rnc</code> , where $\langle \langle \mathit{SIId} \rangle \rangle$ stands for the sub-language identifier. |

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| kohlhase-omdoc_<br>F00342 | 238  | ■ 1.000 | $2\ n$                                                                                            | $2n$                                                                  | so achieve interoperability for $n$ systems with only $2n$ translations instead of                                                                                |
| kohlhase-omdoc_<br>F00343 | 238  | ■ 1.000 | $n^{\{2\}}$                                                                                       | $n^2$                                                                 | $n^2$ . As we have seen in Section 2.1, OPENMATH and Content-MATHML pro-                                                                                          |
| kohlhase-omdoc_<br>F00344 | 240  | ■ 0.970 | $\lambda\mathrm{C\ l\ a\ m}$                                                                      | $\lambda Clam$                                                        | systems $\Omega$ MEGA [BCF <sup>+</sup> 97], INKA [HS96], Pvs [ORS92], $\lambda Clam$ [RSG98b],                                                                   |
| kohlhase-omdoc_<br>F00345 | 240  | ■ 0.836 | $\left[\mathrm{ABI}^{\{+\}}96\right]$                                                             | $[\mathrm{ABI}^+96]$                                                  | TPS <span style="border: 1px solid red;">[ABI<sup>+</sup>96]</span> and CoQ [Tea] see Section 26.18                                                               |
| kohlhase-omdoc_<br>F00346 | 241  | ■ 0.183 | $\mathrm{IAT}_{\{E\}}\ \mathrm{X}$                                                                | $\mathrm{IAT}_E\mathrm{X}$                                            | omdoc2html.xsl for XHTML+MATHML and omdoc2tex.xsl for L <sup>A</sup> T <sub>E</sub> X.                                                                            |
| kohlhase-omdoc_<br>F00347 | 243  | ■ 0.990 | $\mathrm{T}_{\{\mathrm{E}\}}\ \mathrm{X}_{\{\mathrm{MACS}\}}$                                     | $\mathrm{T}_E\mathrm{X}_{\mathrm{MACS}}$                              | plugin for MOZILLA-based browsers, and the plugin for T <sub>E</sub> X <sub>MACS</sub> (Sec-                                                                      |
| kohlhase-omdoc_<br>F00348 | 243  | ■ 1.000 | $\mathrm{ST}_{\{\mathrm{E}\}}\ \mathrm{X}$                                                        | $\mathrm{ST}_E\mathrm{X}$                                             | OMDOC that can be generated by the qmath parser. sT <sub>E</sub> X defines a                                                                                      |
| kohlhase-omdoc_<br>F00349 | 243  | ■ 0.220 | $_{\{\mathrm{E}\}}$                                                                               | $E$                                                                   | XHTML+MATHML or L <sup>A</sup> T <sub>E</sub> X, which can in turn be published on the                                                                            |
| kohlhase-omdoc_<br>F00350 | 246  | ■ 1.000 | $2+2=4$                                                                                           | $2 + 2 = 4$                                                           | Freedom is the freedom to say $\$2+2=4\$$ .                                                                                                                       |
| kohlhase-omdoc_<br>F00351 | 251  | ■ 0.429 | $\$ \ \mathrm{e}^{\wedge\{\mathrm{pi}\}} * \mathrm{i} + 1 = 0 \ \$$                               | $\$e^{\wedge(\mathrm{pi} * \mathrm{i})} + 1 = 0\$$                    | is similar to typing T <sub>E</sub> X or QMATH text: one can type $\$e^{\wedge(\mathrm{pi} * \mathrm{i})} + 1 = 0\$$ and                                          |
| kohlhase-omdoc_<br>F00352 | 251  | ■ 1.000 | $e^{\wedge\{\mathrm{pi}\ i\}} + 1 = 0$                                                            | $e^{\pi i} + 1 = 0$                                                   | get $e^{\pi i} + 1 = 0$ without having to look at the formula editor or use the                                                                                   |
| kohlhase-omdoc_<br>F00353 | 251  | ■ 0.370 | $\{\}^{\{\mathrm{TM}\}}$                                                                          | $\mathrm{TM}$                                                         | “%pi” for MAXIMA, or “Pi” for YACAS, MAPLE <sup>TM</sup> and MATHEMATICA <sup>®</sup> .                                                                           |
| kohlhase-omdoc_<br>F00354 | 251  | ■ 0.370 | $\{\}^{\{\mathrm{R}\}}$                                                                           | $(\mathrm{R})$                                                        | “%pi” for MAXIMA, or “Pi” for YACAS, MAPLE <sup>TM</sup> and MATHEMATICA <sup>®</sup> .                                                                           |
| kohlhase-omdoc_<br>F00355 | 254  | ■ 1.000 | $0$                                                                                               | $O$                                                                   | servers (see [KF01] for details). In particular, an object $O$ residing on an                                                                                     |
| kohlhase-omdoc_<br>F00356 | 254  | ■ 1.000 | $0^{\{\mathrm{prime}\}}$                                                                          | $O'$                                                                  | MBase server $S$ can refer to (or depend on) an object $O'$ residing on a                                                                                         |
| kohlhase-omdoc_<br>F00357 | 254  | ■ 1.000 | $S^{\{\mathrm{prime}\}}$                                                                          | $S'$                                                                  | server $S'$ ; a query to $O$ that needs information about $O'$ will be delegated to                                                                               |
| kohlhase-omdoc_<br>F00358 | 255  | ■ 1.000 | $\mathrm{binom}\{n\}\{k\}, \{\}_n\ C^{\{k\}}, C_{\{k\}}^{\{n\}}$                                  | $\binom{n}{k}, {}_nC^k, C_k^n$                                        | $\binom{n}{k}$ , ${}_nC^k$ , $C_k^n$ , and $C_n^k$ all mean the same thing: $\frac{n!}{k!(n-k)!}$ . In a formula                                                  |
| kohlhase-omdoc_<br>F00359 | 255  | ■ 1.000 | $C_{\{n\}}^{\{k\}}$                                                                               | $C_n^k$                                                               | $\binom{n}{k}$ , ${}_nC^k$ , $C_k^n$ , and $C_n^k$ all mean the same thing: $\frac{n!}{k!(n-k)!}$ . In a formula                                                  |
| kohlhase-omdoc_<br>F00360 | 255  | ■ 1.000 | $\frac{n!}{k!(n-k)!}$                                                                             | $\frac{n!}{k!(n-k)!}$                                                 | $\binom{n}{k}$ , ${}_nC^k$ , $C_k^n$ , and $C_n^k$ all mean the same thing: $\frac{n!}{k!(n-k)!}$ . In a formula                                                  |
| kohlhase-omdoc_<br>F00361 | 255  | ■ 0.999 | $\int f(x)\ d\ x$                                                                                 | $\int f(x)dx$                                                         | jects, e.g. an integral expression of the form $\int f(x)dx$ can mean a Rie-                                                                                      |
| kohlhase-omdoc_<br>F00362 | 255  | ■ 1.000 | $\int f(y)\ d\ y$                                                                                 | $\int f(y)dy$                                                         | stance $\int f(x)dx$ means the same as $\int f(y)dy$ (modulo $\alpha$ -equivalence), so                                                                           |
| kohlhase-omdoc_<br>F00363 | 255  | ■ 1.000 | $s(t)$                                                                                            | $s(t)$                                                                | cal problem such as finding the power of a given signal $s(t)$ . Of course our                                                                                    |
| kohlhase-omdoc_<br>F00364 | 255  | ■ 1.000 | $\int_{\{?\}}^{\{?\}} s^{\{2\}}(t)\ d\ t$                                                         | $\int_{\{?\}}^{\{?\}} s^2(t)dt$                                       | MATHWEBSEARCH to search for something that looks like $\int_{\{?\}}^{\{?\}} s^2(t)dt$ (for                                                                        |
| kohlhase-omdoc_<br>F00365 | 255  | ■ 1.000 | $\frac{1}{T} \int_{\{0\}}^{\{T\}} s^{\{2\}}(t)\ d\ t = \Sigma_{k=-\infty}^{\infty}  c_{\{k\}} ^2$ | $\frac{1}{T} \int_0^T s^2(t)dt = \Sigma_{k=-\infty}^{\infty}  c_k ^2$ | $\frac{1}{T} \int_0^T s^2(t)dt = \Sigma_{k=-\infty}^{\infty}  c_k ^2$ where $c_k$ <span style="border: 1px solid red;">are the Fourier coefficients of the</span> |
| kohlhase-omdoc_<br>F00366 | 255  | ■ 1.000 | $c_{\{k\}}$                                                                                       | $c_k$                                                                 | $\frac{1}{T} \int_0^T s^2(t)dt = \Sigma_{k=-\infty}^{\infty}  c_k ^2$ where $c_k$ are the Fourier coefficients of the                                             |
| kohlhase-omdoc_<br>F00367 | 257  | ■ 1.000 | $A=B$                                                                                             | $A = B$                                                               | While of searching expressions of the form $A = B$ , we might like to find                                                                                        |

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|-----------------------|------|-------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| kohlhase-omdoc_F00368 | 257  | 0.737 | B=A                                                                                                       | $B = A$                                                                     | occurrences of $B = A$ as well. At this point the <code>mq:anyorder</code> attribute comes                                                 |
| kohlhase-omdoc_F00369 | 258  | 0.994 | $\frac{1}{T} \int_0^T s^2(x) dx = \Sigma_{k=-\infty}^{\infty}  c_k ^2$                                    | $\frac{1}{T} \int_0^T s^2(x) dx = \Sigma_{k=-\infty}^{\infty}  c_k ^2$      | the variant $\frac{1}{T} \int_0^T s^2(x) dx = \Sigma_{k=-\infty}^{\infty}  c_k ^2$ .                                                       |
| kohlhase-omdoc_F00370 | 259  | 0.319 | $\left. \right\}^{+} \text{03, } \mathrm{KBLL} \{ \}^{+} \text{04, } \mathrm{KBKB}^{+} \text{04} \right]$ | $^{+} \text{03, KBLL}^{+} \text{04, KBKB}^{+} \text{04}]$                   | (see [KBHL <sup>+03</sup> , KBLL <sup>+04</sup> , KBKB <sup>+04</sup> ]).                                                                  |
| kohlhase-omdoc_F00371 | 261  | 1.000 | T                                                                                                         | $T$                                                                         | entities, such that (for a theorem $T$ a proof $P$ , and sections $A, B$ ):                                                                |
| kohlhase-omdoc_F00372 | 261  | 1.000 | A, B                                                                                                      | $A, B$                                                                      | entities, such that (for a theorem $T$ a proof $P$ , and sections $A, B$ ):                                                                |
| kohlhase-omdoc_F00373 | 261  | 0.533 | P & A                                                                                                     | $P \& A$                                                                    | $B$ contains $P$ & $A$ contains $T$ & $P$ proves $T \Rightarrow B$ reliesOn $A$ .                                                          |
| kohlhase-omdoc_F00374 | 261  | 0.533 | T & P                                                                                                     | $T \& P$                                                                    | $B$ contains $P$ & $A$ contains $T$ & $P$ proves $T \Rightarrow B$ reliesOn $A$ .                                                          |
| kohlhase-omdoc_F00375 | 261  | 0.533 | T $\Rightarrow$ B                                                                                         | $T \Rightarrow B$                                                           | $B$ contains $P$ & $A$ contains $T$ & $P$ proves $T \Rightarrow B$ reliesOn $A$ .                                                          |
| kohlhase-omdoc_F00376 | 264  | 0.997 | $\mathbb{F}$                                                                                              | $\mathbb{F}$                                                                | ematical objects ‘live’. For instance, when a chapter begins with “Let $\mathbb{F}$ be                                                     |
| kohlhase-omdoc_F00377 | 264  | 1.000 | $\frac{3}{4} / \frac{2}{3} / \frac{3}{5}$                                                                 | $\frac{3/4 + 2/3}{5}$                                                       | fraction display. In $\frac{3/4+2/3}{5}$ we have used them both.                                                                           |
| kohlhase-omdoc_F00378 | 264  | 0.998 | $3, 4 \in \mathbb{Z}$                                                                                     | $3, 4 \in \mathbb{Z}$                                                       | $3, 4 \in \mathbb{Z}$ . The corresponding OPENMATH expression would be the equivalent                                                      |
| kohlhase-omdoc_F00379 | 264  | 1.000 | $3 \in \mathbb{Z} \wedge 4 \in \mathbb{Z}$                                                                | $3 \in \mathbb{Z} \wedge 4 \in \mathbb{Z}$                                  | of $3 \in \mathbb{Z} \wedge 4 \in \mathbb{Z}$ , but our OPENMATH statement reads that the sequence                                         |
| kohlhase-omdoc_F00380 | 264  | 0.998 | $\mathbb{Z}$                                                                                              | $\mathbb{Z}$                                                                | $3, 4$ belongs to $\mathbb{Z}$ . So here, the semantics of the element-of symbol has been                                                  |
| kohlhase-omdoc_F00381 | 268  | 0.608 | $\left[ \mathrm{MGH}^{+} \text{05} \right]$                                                               | $[\mathrm{MGH}^{+} \text{05}]$                                              | and tools $[\mathrm{MGH}^{+} \text{05}]$ .                                                                                                 |
| kohlhase-omdoc_F00382 | 269  | 0.995 | $\left[ \mathrm{KAB}^{+} \text{04} \right]$                                                               | $[\mathrm{KAB}^{+} \text{04}]$                                              | ment (PISA) $[\mathrm{KAB}^{+} \text{04}]$ and National Council of Teachers of Mathematics                                                 |
| kohlhase-omdoc_F00383 | 273  | 0.996 | $10^{\prime} \text{000}$                                                                                  | $10'000$                                                                    | To date, more than $10'000$ <i>items</i> of OMDoc content has been written                                                                 |
| kohlhase-omdoc_F00384 | 273  | 1.000 | $1, \ldots, k, \ldots, n$                                                                                 | $1, \dots, k, \dots, n$                                                     | age of the ellipsis in such a formula as $\underline{1, \dots, k, \dots, n}$ is a simple example of                                        |
| kohlhase-omdoc_F00385 | 275  | 0.998 | $\left[ \mathrm{SBB}^{+} \text{06} \right]$                                                               | $[\mathrm{SBB}^{+} \text{06}]$                                              | learning $[\mathrm{SBB}^{+} \text{06}]$ .                                                                                                  |
| kohlhase-omdoc_F00386 | 279  | 1.000 | $\{ \}^{\sim}$                                                                                            | $\sim$                                                                      | <a href="http://www.cs.nott.ac.uk/~lad/research/">http://www.cs.nott.ac.uk/~lad/research/</a>                                              |
| kohlhase-omdoc_F00388 | 282  | 0.757 | $\{ \}^{+} \text{00}$                                                                                     | $^{+} \text{00}$                                                            | SL $[\mathrm{AHL}^{+} \text{00}]$ . All these specification languages provide constructs similar                                           |
| kohlhase-omdoc_F00389 | 286  | 0.830 | $\left[ \mathrm{SHB}^{+} \text{99} \right]$                                                               | $[\mathrm{SHB}^{+} \text{99}]$                                              | of the GUI, shared with the $\Omega$ MEGA-system $[\mathrm{SHB}^{+} \text{99}]$ , are written                                              |
| kohlhase-omdoc_F00390 | 295  | 0.465 | $\mathrm{St}_{\mathrm{E}} \{ \mathrm{E} \} \mathrm{X}$                                                    | $\mathrm{St}_{\mathrm{E}} \mathrm{X}$                                       | We present $\mathrm{sT}_{\mathrm{E}} \mathrm{X}$ (Semantic $\mathrm{T}_{\mathrm{E}} \mathrm{X}$ ) a collection of macro packages for       |
| kohlhase-omdoc_F00391 | 295  | 0.962 | $\mathrm{ST}_{\mathrm{E}} \{ \mathrm{E} \}^{\mathrm{X}} \{ \mathrm{X} \}$                                 | $\mathrm{ST}_{\mathrm{E}}^{\mathrm{X}} \{ \mathrm{X} \}$                    | documents can be authored and maintained in $\mathrm{sT}_{\mathrm{E}} \mathrm{X}$ using a simple text ed-                                  |
| kohlhase-omdoc_F00392 | 295  | 0.580 | $\mathrm{T}_{\mathrm{E}} \mathrm{X} / \mathrm{IAT}_{\mathrm{E}} \mathrm{X}$                               | $\mathrm{T}_{\mathrm{E}} \mathrm{X} / \mathrm{IAT}_{\mathrm{E}} \mathrm{X}$ | – The $\mathrm{T}_{\mathrm{E}} \mathrm{X} / \mathrm{IAT}_{\mathrm{E}} \mathrm{X}$ syntax is much more compact than OMDoc, and if           |
| kohlhase-omdoc_F00393 | 297  | 0.713 | $\binom{n}{k}, {}_nC^k, \mathcal{C}_k^n$                                                                  | $\binom{n}{k}, {}_nC^k, \mathcal{C}_k^n$                                    | tations” <sup>19</sup> for binomial coefficients: $\binom{n}{k}$ , ${}_nC^k$ , $\mathcal{C}_k^n$ , and $\mathcal{C}_n^k$ all mean the same |
| kohlhase-omdoc_F00394 | 297  | 0.713 | $\mathcal{C}_n^k$                                                                                         | $\mathcal{C}_n^k$                                                           | tations” <sup>19</sup> for binomial coefficients: $\binom{n}{k}$ , ${}_nC^k$ , $\mathcal{C}_k^n$ , and $\mathcal{C}_n^k$ all mean the same |
| kohlhase-omdoc_F00395 | 297  | 1.000 | $\sum_{k=1}^{\infty} x^k$                                                                                 | $\sum_{k=1}^{\infty} x^k$                                                   | for the mathematical expression $\sum_{k=1}^{\infty} x^k$ . In the first form, we specify that we                                          |

| Identifier            | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                   | Rendered                                                                                                                                                      | Image                                                                                                                                                                                                                                                 |
|-----------------------|------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| kohlhase-omdoc_F00396 | 297  | ■ 0.943 | $\infty$                                                                                                                                                                                                                                                                                                                                                                       | $\infty$                                                                                                                                                      | (i) the bound variable $k$ (ii) the number 1 (iii) $\infty$ (iv) $\backslash\mathrm{Cexp}\{x\}k$ (i.e. $x$ to                                                                                                                                         |
| kohlhase-omdoc_F00397 | 297  | ■ 0.870 | $k=1$                                                                                                                                                                                                                                                                                                                                                                          | $k = 1$                                                                                                                                                       | capital sigma character ( $\Sigma$ ) whose subscript is the equation $k=$ and whose                                                                                                                                                                   |
| kohlhase-omdoc_F00398 | 298  | ■ 0.996 | $\sum_{i=1}^n x^i$                                                                                                                                                                                                                                                                                                                                                             | $\sum_{i=1}^n x^i$                                                                                                                                            | an expression like $\sum_{i=1}^n x^i$ as $\backslash\mathrm{Sumfromto}\{i\}1n\{2i-1\}$ . In this example we                                                                                                                                           |
| kohlhase-omdoc_F00399 | 298  | ■ 1.000 | $\backslash\mathrm{arbitrary}n$                                                                                                                                                                                                                                                                                                                                                | (not rendered)                                                                                                                                                | What is the sum of the first $\backslash\mathrm{arbitrary}n$ odd numbers, i.e.                                                                                                                                                                        |
| kohlhase-omdoc_F00400 | 298  | ■ 1.000 | $\backslash\mathrm{Sumfromto}\{i\}1\backslash\mathrm{arbitrary}n\{2i-1\}?$                                                                                                                                                                                                                                                                                                     | (not rendered)                                                                                                                                                | $\backslash\mathrm{Sumfromto}\{i\}1\backslash\mathrm{arbitrary}n\{2i-1\}?\$$                                                                                                                                                                          |
| kohlhase-omdoc_F00401 | 298  | ■ 0.998 | $\sum_{i=1}^n 2i - 1$                                                                                                                                                                                                                                                                                                                                                          | $\sum_{i=1}^n 2i - 1$                                                                                                                                         | <u>What is the sum of the first <math>n</math> odd numbers, i.e. <math>\sum_{i=1}^n 2i - 1</math>?</u>                                                                                                                                                |
| kohlhase-omdoc_F00402 | 298  | ■ 0.660 | $\backslash\mathrm{mathrm}\{S\}_{\backslash\mathrm{mathrm}\{E\}} \backslash\mathrm{mathrm}\{X\}$                                                                                                                                                                                                                                                                               | $S_{\mathrm{E}}X$                                                                                                                                             | The $\mathrm{S}\mathrm{I}\mathrm{E}\mathrm{X}$ macro packages have been validated together with a case                                                                                                                                                |
| kohlhase-omdoc_F00403 | 303  | ■ 0.999 | $a x^2 + b x + c$                                                                                                                                                                                                                                                                                                                                                              | $ax^2 + bx + c$                                                                                                                                               | mial expression $ax^2 + bx + c$ whose internal representation in MATHEMATICA <sup>®</sup>                                                                                                                                                             |
| kohlhase-omdoc_F00405 | 306  | ■ 1.000 | $\rho_*$                                                                                                                                                                                                                                                                                                                                                                       | $\rho_*$                                                                                                                                                      | inclusions $\rho_*$ that interpret the system concepts in terms of <code>mbase:sets</code> .                                                                                                                                                          |
| kohlhase-omdoc_F00406 | 306  | ■ 1.000 | $\rho_*^{-1}$                                                                                                                                                                                                                                                                                                                                                                  | $\rho_*^{-1}$                                                                                                                                                 | the integration theory, using the partial inverse $\rho_*^{-1}$ of $\rho_*$ . For an integration                                                                                                                                                      |
| kohlhase-omdoc_F00407 | 308  | ■ 1.000 | $\backslash\mathrm{mathcal}\{S\}=(\backslash\mathrm{mathcal}\{L\}, \backslash\mathrm{mathcal}\{C\})$                                                                                                                                                                                                                                                                           | $\mathcal{S} = (\mathcal{L}, \mathcal{C})$                                                                                                                    | <b>Definition 26.18.1.</b> A <i>logical system</i> $\mathcal{S} = (\mathcal{L}, \mathcal{C})$ consists of a language $\mathcal{L}$                                                                                                                    |
| kohlhase-omdoc_F00408 | 308  | ■ 1.000 | $\backslash\mathrm{mathcal}\{L\}$                                                                                                                                                                                                                                                                                                                                              | $\mathcal{L}$                                                                                                                                                 | <b>Definition 26.18.1.</b> A <i>logical system</i> $\mathcal{S} = (\mathcal{L}, \mathcal{C})$ consists of a language $\mathcal{L}$                                                                                                                    |
| kohlhase-omdoc_F00409 | 308  | ■ 0.999 | $\backslash\mathrm{mathcal}\{H\}$                                                                                                                                                                                                                                                                                                                                              | $\mathcal{H}$                                                                                                                                                 | rules). A calculus gives us a notion of a $\mathcal{C}$ -derivation of $\mathbf{A}$ from $\mathcal{H}$ , which                                                                                                                                        |
| kohlhase-omdoc_F00410 | 309  | ■ 1.000 | $\backslash\mathrm{mathcal}\{D\}: \backslash\mathrm{mathcal}\{H\} \vdash_{\mathcal{C}} \backslash\mathrm{mathbf}\{A\}$                                                                                                                                                                                                                                                         | $\mathcal{D} : \mathcal{H} \vdash_{\mathcal{C}} \mathbf{A}$                                                                                                   | we will denote by $\mathcal{D} : \mathcal{H} \vdash_{\mathcal{C}} \mathbf{A}$ . Let $\mathcal{S}$ and $\mathcal{S}'$ be logical systems, then a <i>logic</i>                                                                                          |
| kohlhase-omdoc_F00411 | 309  | ■ 1.000 | $\backslash\mathrm{mathcal}\{S\}^{\backslash\mathrm{prime}}$                                                                                                                                                                                                                                                                                                                   | $\mathcal{S}'$                                                                                                                                                | we will denote by $\mathcal{D} : \mathcal{H} \vdash_{\mathcal{C}} \mathbf{A}$ . Let $\mathcal{S}$ and $\mathcal{S}'$ be logical systems, then a <i>logic</i>                                                                                          |
| kohlhase-omdoc_F00412 | 309  | ■ 1.000 | $\backslash\mathrm{mathcal}\{F\}: \backslash\mathrm{mathcal}\{S\} \rightarrow \backslash\mathrm{mathcal}\{S\}^{\backslash\mathrm{prime}}$                                                                                                                                                                                                                                      | $\mathcal{F} : \mathcal{S} \rightarrow \mathcal{S}'$                                                                                                          | <i>morphism</i> $\mathcal{F} : \mathcal{S} \rightarrow \mathcal{S}'$ consists of a <i>language morphism</i> $\mathcal{F}^{\mathcal{L}} : \mathcal{L} \rightarrow \mathcal{L}'$ and                                                                    |
| kohlhase-omdoc_F00413 | 309  | ■ 1.000 | $\backslash\mathrm{mathcal}\{F\}^{\backslash\mathrm{mathcal}\{L\}}: \backslash\mathrm{mathcal}\{L\} \rightarrow \backslash\mathrm{mathcal}\{L\}^{\backslash\mathrm{prime}}$                                                                                                                                                                                                    | $\mathcal{F}^{\mathcal{L}} : \mathcal{L} \rightarrow \mathcal{L}'$                                                                                            | <i>morphism</i> $\mathcal{F} : \mathcal{S} \rightarrow \mathcal{S}'$ consists of a <i>language morphism</i> $\mathcal{F}^{\mathcal{L}} : \mathcal{L} \rightarrow \mathcal{L}'$ and                                                                    |
| kohlhase-omdoc_F00414 | 309  | ■ 1.000 | $\backslash\mathrm{mathcal}\{F\}^{\backslash\mathrm{mathcal}\{D\}}$                                                                                                                                                                                                                                                                                                            | $\mathcal{F}^{\mathcal{D}}$                                                                                                                                   | a <i>calculus morphism</i> $\mathcal{F}^{\mathcal{D}}$ from $\mathcal{C}$ -derivations to $\mathcal{C}'$ -derivations, such that                                                                                                                      |
| kohlhase-omdoc_F00415 | 309  | ■ 1.000 | $\backslash\mathrm{mathcal}\{C\}^{\backslash\mathrm{prime}}$                                                                                                                                                                                                                                                                                                                   | $\mathcal{C}'$                                                                                                                                                | a <i>calculus morphism</i> $\mathcal{F}^{\mathcal{D}}$ from $\mathcal{C}$ -derivations to $\mathcal{C}'$ -derivations, such that                                                                                                                      |
| kohlhase-omdoc_F00416 | 309  | ■ 0.999 | $\backslash\mathrm{mathcal}\{F\}^{\backslash\mathrm{mathcal}\{D\}}(\backslash\mathrm{mathcal}\{D\}): \backslash\mathrm{mathcal}\{F\}^{\backslash\mathrm{mathcal}\{L\}}(\backslash\mathrm{mathcal}\{H\}) \vdash_{\backslash\mathrm{mathcal}\{C\}^{\backslash\mathrm{prime}}} \backslash\mathrm{mathcal}\{F\}^{\backslash\mathrm{mathcal}\{L\}}(\backslash\mathrm{mathbf}\{A\})$ | $\mathcal{F}^{\mathcal{D}}(\mathcal{D}) : \mathcal{F}^{\mathcal{L}}(\mathcal{H}) \vdash_{\mathcal{C}'} \mathcal{F}^{\mathcal{L}}(\mathbf{A})$                 | for any $\mathcal{C}$ -derivation $\mathcal{D} : \mathcal{H} \vdash_{\mathcal{C}} \mathbf{A}$ we have $\mathcal{F}^{\mathcal{D}}(\mathcal{D}) : \mathcal{F}^{\mathcal{L}}(\mathcal{H}) \vdash_{\mathcal{C}'} \mathcal{F}^{\mathcal{L}}(\mathbf{A})$ . |
| kohlhase-omdoc_F00417 | 309  | ■ 1.000 | $\backslash\mathrm{mathcal}\{R\}$                                                                                                                                                                                                                                                                                                                                              | $\mathcal{R}$                                                                                                                                                 | instance the morphism $\mathcal{R}$ from sorted first-order logic ( $\mathcal{S}\mathrm{FOL}$ ) to unsorted                                                                                                                                           |
| kohlhase-omdoc_F00418 | 309  | ■ 1.000 | $(\backslash\mathrm{mathbb}\{S\} \text{ F O L})$                                                                                                                                                                                                                                                                                                                               | $(\mathcal{S}\mathrm{FOL})$                                                                                                                                   | instance the morphism $\mathcal{R}$ from sorted first-order logic ( $\mathcal{S}\mathrm{FOL}$ ) to unsorted                                                                                                                                           |
| kohlhase-omdoc_F00419 | 309  | ■ 1.000 | $(\text{F O L})$                                                                                                                                                                                                                                                                                                                                                               | $(\mathrm{FOL})$                                                                                                                                              | first-order logic ( $\mathrm{FOL}$ ). For every sorted constant $\mathcal{R}$ introduces an axiom                                                                                                                                                     |
| kohlhase-omdoc_F00420 | 309  | ■ 1.000 | $\backslash\mathrm{mathcal}\{R\}([+: \backslash\mathrm{mathbb}\{N\} \rightarrow \backslash\mathrm{mathbb}\{N\} \rightarrow \backslash\mathrm{mathbb}\{N\}) = \forall X, Y. \backslash\mathrm{mathbb}\{N\}(X) \wedge \backslash\mathrm{mathbb}\{N\}(Y) \Rightarrow \backslash\mathrm{mathbb}\{N\}(X+Y)$                                                                         | $\mathcal{R}([+: \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N}]) = \forall X, Y. \mathbb{N}(X) \wedge \mathbb{N}(Y) \Rightarrow \mathbb{N}(X + Y)$ | e.g. $\mathcal{R}([+: \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N}]) = \forall X, Y. \mathbb{N}(X) \wedge \mathbb{N}(Y) \Rightarrow \mathbb{N}(X + Y)$ . On formu-                                                                        |
| kohlhase-omdoc_F00421 | 309  | ■ 0.999 | $\backslash\mathrm{mathcal}\{R\}(\backslash\mathrm{forall} X_{\backslash\mathrm{mathbb}\{B\}} . \backslash\mathrm{mathbf}\{A\} \rightarrow \backslash\mathrm{forall} X . \backslash\mathrm{mathbb}\{B\}(X) \rightarrow \backslash\mathrm{mathcal}\{R\}(\backslash\mathrm{mathbf}\{A\}))$                                                                                       | $\mathcal{R}(\forall X_{\mathbb{B}}. \mathbf{A}) = \forall X. \mathbb{B}(X) \Rightarrow \mathcal{R}(\mathbf{A})$                                              | predicates, e.g. $\mathcal{R}(\forall X_{\mathbb{B}}. \mathbf{A}) = \forall X. \mathbb{B}(X) \Rightarrow \mathcal{R}(\mathbf{A})$ . Finally, for proofs we have                                                                                       |
| kohlhase-omdoc_F00422 | 309  | ■ 1.000 | $\backslash\mathrm{mathbb}\{A\}, \backslash\mathrm{mathbb}\{B\}, \backslash\mathrm{ldots}$                                                                                                                                                                                                                                                                                     | $\mathbb{A}, \mathbb{B}, \dots$                                                                                                                               | the correspondence given in Figure 26.26, where $\mathbb{A}, \mathbb{B}, \dots$ are sort symbols.                                                                                                                                                     |

| Identifier                | Page | Conf.   | LaTeX source                                                                                                                          | Rendered                                                                     | Image                                                                                                                                                                                             |
|---------------------------|------|---------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| kohlhase-omdoc_<br>F00423 | 309  | ■ 0.989 | $\mathbb{S} \text{ F O L}$                                                                                                            | $\mathcal{S}FOL$                                                             | simply be encoded as a theory inclusion from $\mathcal{S}FOL$ to $FOL$ mapping $\mathcal{S}FOL$                                                                                                   |
| kohlhase-omdoc_<br>F00424 | 309  | ■ 0.989 | $\text{F O L}$                                                                                                                        | $FOL$                                                                        | simply be encoded as a theory inclusion from $\mathcal{S}FOL$ to $FOL$ mapping $\mathcal{S}FOL$                                                                                                   |
| kohlhase-omdoc_<br>F00425 | 310  | ■ 0.752 | $\left[\mathrm{SBB}^{\{+\}} \text{ 02}\right]$                                                                                        | $\left[\mathrm{SBB}^{+02}\right]$                                            | $\Omega\text{MEGA}$ [SBB <sup>+</sup> 02, SBA05]. Generally, we aim at an approach that is not de-                                                                                                |
| kohlhase-omdoc_<br>F00426 | 310  | ■ 0.975 | $\mathrm{T}_{\{\mathrm{E}\}} \mathrm{X}_{\{\text{MACS}\}} [\mathrm{dH} \text{ 01}]$                                                   | $\mathrm{T}_{\mathrm{E}}\mathrm{X}_{\mathrm{MACS}} [\mathrm{dH}01]$          | $\mathrm{T}_{\mathrm{E}}\mathrm{X}_{\mathrm{MACS}} [\mathrm{dH}01]$ is a scientific WYSIWYG text editor that provides                                                                             |
| kohlhase-omdoc_<br>F00427 | 312  | ■ 1.000 | $\mathrm{T} + \mathrm{S}$                                                                                                             | $T + S$                                                                      | we define a <b>semantic document format</b> $T + S$ as a document format still                                                                                                                    |
| kohlhase-omdoc_<br>F00428 | 312  | ■ 1.000 | $\mathrm{S}(\mathrm{P})$                                                                                                              | $S(P)$                                                                       | defining a fixed language $S$ , we define a language $S(P)$ parametrized over a                                                                                                                   |
| kohlhase-omdoc_<br>F00429 | 312  | ■ 1.000 | $\mathrm{S}(\mathrm{C})$                                                                                                              | $S(C)$                                                                       | language $P$ for proofs and define the document format based on $S(C)$ , where                                                                                                                    |
| kohlhase-omdoc_<br>F00430 | 312  | ■ 1.000 | $\mathrm{C}$                                                                                                                          | $C$                                                                          | $C$ denotes the proof language of CORE. This format supports the <i>static repre-</i>                                                                                                             |
| kohlhase-omdoc_<br>F00431 | 312  | ■ 0.993 | $\mathrm{T} + \mathrm{S}(\mathrm{C})$                                                                                                 | $T + S(C)$                                                                   | The $\mathrm{T}_{\mathrm{E}}\mathrm{X}_{\mathrm{MACS}}$ document $T + S(C)$ and the pure semantic representation                                                                                  |
| kohlhase-omdoc_<br>F00432 | 312  | ■ 0.993 | $\mathrm{t} \text{ s}_{\{i\}}$                                                                                                        | $ts_i$                                                                       | $\mathrm{T}_{\mathrm{E}}\mathrm{X}_{\mathrm{MACS}}$ documents. The differences between two versions $ts_i$ and $ts_{i+1}$ of                                                                      |
| kohlhase-omdoc_<br>F00433 | 312  | ■ 0.993 | $\mathrm{t} \text{ s}_{\{i+1\}}$                                                                                                      | $ts_{i+1}$                                                                   | $\mathrm{T}_{\mathrm{E}}\mathrm{X}_{\mathrm{MACS}}$ documents. The differences between two versions $ts_i$ and $ts_{i+1}$ of                                                                      |
| kohlhase-omdoc_<br>F00434 | 312  | ■ 1.000 | $\mathrm{s}_{\{i\}}$                                                                                                                  | $s_i$                                                                        | corresponding document $s_i$ in $S(C)$ for $ts_i$ , such that the application of $p$                                                                                                              |
| kohlhase-omdoc_<br>F00435 | 312  | ■ 1.000 | $\mathrm{s}_{\{i+1\}}$                                                                                                                | $s_{i+1}$                                                                    | to $s_i$ results in $s_{i+1}$ which corresponds to $ts_{i+1}$ . An analogous diff/patch                                                                                                           |
| kohlhase-omdoc_<br>F00436 | 313  | ■ 1.000 | $\mathrm{T} + \mathrm{S}(\mathrm{C}) + \mathrm{M}$                                                                                    | $T + S(C) + M$                                                               | extended to $T + S(C) + M$ , where the menus can be attached to arbitrary                                                                                                                         |
| kohlhase-omdoc_<br>F00437 | 313  | ■ 1.000 | $\mathrm{S}(\mathrm{C}) + \mathrm{M}$                                                                                                 | $S(C) + M$                                                                   | and $S(C) + M$ via the diff/patch mechanism. Note that this includes also                                                                                                                         |
| kohlhase-omdoc_<br>F00438 | 314  | ■ 1.000 | $\mathcal{F} \mathcal{P}$                                                                                                             | $\mathcal{FP}$                                                               | gramming language $\mathcal{FP}$ . The system has been developed since 1998 at                                                                                                                    |
| kohlhase-omdoc_<br>F00439 | 316  | ■ 0.537 | $h_{\{1\}}, \ldots h_{\{n\}}, \forall \ldots i h_{\{1\}}, \ldots, \forall \ldots i h_l \vdash$<br>$\forall \ldots i h_{\{l\}} \vdash$ | $h_1, \ldots h_n, \forall \ldots i h_1, \ldots, \forall \ldots i h_l \vdash$ | with a sequence of the form <span style="border: 1px solid red; padding: 2px;"><math>h_1, \ldots h_n, \forall \ldots i h_1, \ldots, \forall \ldots i h_l \vdash</math></span> <i>goal</i> whereas |
| kohlhase-omdoc_<br>F00440 | 316  | ■ 1.000 | $h_{\{i\}}$                                                                                                                           | $h_i$                                                                        | the expressions $h_i$ are the hypotheses, the expressions $\forall \ldots i h_k$ are the in-                                                                                                      |
| kohlhase-omdoc_<br>F00441 | 316  | ■ 1.000 | $\forall \ldots i h_{\{k\}}$                                                                                                          | $\forall \ldots i h_k$                                                       | the expressions $h_i$ are the hypotheses, the expressions $\forall \ldots i h_k$ are the in-                                                                                                      |
| kohlhase-omdoc_<br>F00445 | 374  | ■ 0.966 | $\{ \}^{\{+\}} \text{ 02}$                                                                                                            | $^{+02}$                                                                     | [BCD <sup>+</sup> 02] R. Bradford, R. M. Corless, J. H. Davenport, D. J. Jeffrey, and S. M.                                                                                                       |
| kohlhase-omdoc_<br>F00446 | 375  | ■ 1.000 | $\bar{\lambda} \mu \bar{\mu}$                                                                                                         | $\bar{\lambda} \mu \bar{\mu}$                                                | [Coe05] Claudio Sacerdoti Coen. Explanation in natural language of $\bar{\lambda} \mu \bar{\mu}$ -terms.                                                                                          |
| kohlhase-omdoc_<br>F00447 | 377  | ■ 0.891 | $\text{H O L} - \text{A}$                                                                                                             | $HOL - A$                                                                    | [GM93] M. J. C. Gordon and T. F. Melham. <i>Introduction to HOL – A theorem</i>                                                                                                                   |
| kohlhase-omdoc_<br>F00448 | 379  | ■ 1.000 | $\mathrm{T}_{\{\mathrm{E}\}} \text{ X b o o k}$                                                                                       | $T_{\mathrm{E}}\mathrm{X}book$                                               | [Knu84] Donald E. Knuth. <i>The <math>T_{\mathrm{E}}\mathrm{X}book</math></i> . Addison Wesley, 1984.                                                                                             |
| kohlhase-omdoc_<br>F00449 | 380  | ■ 0.807 | $\{ \}^{\{\mathrm{A} \text{ T}\}} \mathrm{T}_{\{\mathrm{E}\}} \text{ X}$                                                              | $^{\mathrm{AT}}T_{\mathrm{E}}\mathrm{X}$                                     | [Lam94] Leslie Lamport. <i><math>\mathcal{B}T_{\mathrm{E}}\mathrm{X}</math>: A Document Preparation System, 2/e</i> . Addison                                                                     |
| kohlhase-omdoc_<br>F00450 | 383  | ■ 0.842 | $\lambda \text{ c l a m}$                                                                                                             | $\lambda clam$                                                               | scription: Proof planning in higher-order logic with $\lambda clam$ . In Claude Kirchner                                                                                                          |
| kohlhase-omdoc_<br>F00451 | 383  | ■ 0.990 | $\{ \}^{\{+\}} \text{ 06}$                                                                                                            | $^{+06}$                                                                     | [SBB <sup>+</sup> 06] Sebastian Schaffert, Diana Bischof, Tobias Bürger, Andreas Gruber,                                                                                                          |
| kohlhase-omdoc_<br>F00452 | 384  | ■ 0.826 | $\{ \}_{\{i\}}$                                                                                                                       | $\iota$                                                                      | [Sci] Design Science. Mathplayer $\displaystyle \mathit{math}$ in your browser $\iota$ . web page at                                                                                              |
| kohlhase-omdoc_<br>F00453 | 384  | ■ 0.614 | $\mathcal{L} \mathcal{O} \mathcal{U} \mathcal{I}$                                                                                     | $\mathcal{LOUI}$                                                             | Konrad, Andreas Meier, Erica Melis, Martin Pollet, and Volker Sorge. $\mathcal{LOUI}$ :                                                                                                           |

| Identifier            | Page | Conf.   | LaTeX source                                             | Rendered                                                 | Image                                                                                                               |
|-----------------------|------|---------|----------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| kohlhase-omdoc_F00454 | 384  | ■ 0.952 | $\mathcal{U}$                                            | $\mathcal{U}$                                            | <i>Lovely <math>\Omega</math>MEGA User Interface. Formal Aspects of Computing, 11:326–342, 1999.</i>                |
| kohlhase-omdoc_F00455 | 384  | ■ 0.952 | $\mathcal{I}$                                            | $\mathcal{I}$                                            | <i>Lovely <math>\Omega</math>MEGA User Interface. Formal Aspects of Computing, 11:326–342, 1999.</i>                |
| kohlhase-omdoc_F00456 | 386  | ■ 0.604 | X I I I, 23,25,239,285,304                               | $XIII$ , 23, 25, 239, 285, 304                           | $\Omega$ MEGA <span style="border: 1px solid red; padding: 0 2px;"><math>XIII</math>, 23, 25, 239, 285, 304,</span> |
| kohlhase-omdoc_F00457 | 386  | ■ 1.000 | $\bar{\lambda} \mu E \mu$                                | $\bar{\lambda} \mu E \mu$                                | $\bar{\lambda} \mu E \mu$                                                                                           |
| kohlhase-omdoc_F00458 | 421  | ■ 0.942 | $\mathrm{T}_{\mathrm{E}} \mathrm{X}_{\mathrm{MACS}} 309$ | $\mathrm{T}_{\mathrm{E}} \mathrm{X}_{\mathrm{MACS}} 309$ | $\mathrm{T}_{\mathrm{E}} \mathrm{X}_{\mathrm{MACS}} 309$                                                            |